A Capstone Project presented to the College of Information Technology Education PHINMA University of Iloilo

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Design and Development of an Integrated Study Hub Management System for WS Students & Professionals Lounge

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Science in Information Technology

System Development

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Abstract

The rapid expansion of shared workspaces has increased the operational demand for efficient facility management systems. The WS Students & Professionals Lounge – La Paz Branch currently manages its daily transactions through manual workflows, paper logbooks, decentralized spreadsheets, and third-party social media platforms. This fragmented operational setup results in structural vulnerabilities, including resource double-bookings, prolonged customer waiting times, data entry errors, and financial tracking discrepancies. To mitigate these administrative bottlenecks, this study developed the Integrated Study Hub Management System, a centralized, cross-platform web application designed to optimize back-office utilities and enhance user service quality. Utilizing an agile development methodology, the system was engineered with six primary modules: Reservation Management, Membership and Time Tracking, Payment Billing, Sales Recording and Reporting, Admin and Staff Dashboards, and Role-Based Access Control (RBAC).

The technical architecture is built on a web-based framework supported by a relational MySQL database engine, ensuring secure data synchronization and eliminating data fragmentation. Evaluation metrics indicate that the automation of elapsed-time calculations and live space allocation rules effectively eradicates transcription errors, secures historical financial ledgers, and minimizes context-switching for frontline desk personnel. Ultimately, the integration of this system converts manual front-desk friction into actionable business intelligence, providing a secure, transparent, and frictionless facility environment for both administrative personnel and client.

CHAPTER 1

1.1 Introduction to the Study

In the modern digital era, the operational efficiency of service-oriented enterprises relies heavily on the integration of centralized software architectures that mitigate manual friction and eliminate data fragmentation. Shared workspace facilities, such as study hubs and co-working lounges, process complex daily transaction streams that demand high computational precision. These transactions include real-time space scheduling, user subscription authentication, and dynamic, time-based financial billing. When an enterprise manages these highly volatile parameters using non-integrated, manual procedures, its operational framework becomes fundamentally vulnerable to processing delays, human error, and structural data silos. Consequently, modern organizational management dictates that transitioning to an automated, centralized platform is an operational necessity to preserve data integrity and optimize service delivery.

The WS Students & Professionals Lounge – La Paz Branch currently conducts its business functions through a fragmented combination of manual, paper-reliant processes, localized spreadsheet applications, and third-party social media utilities. Specifically, the establishment utilizes Facebook and Facebook Messenger as its primary communication interfaces for coordinating customer reservations and handling client availability inquiries. Frontline desk personnel must manually parse incoming chat logs, verify physical availability, and retrospectively transcribe these transactional details into decentralized paper ledgers or detached spreadsheets. This disconnected manual workflow demands significant administrative labor and creates a highly volatile operational environment where data synchronization cannot be guaranteed in real time.

Additionally, the lounge's customer registration and long-term membership management follow an identical manual methodology. New clients are registered via handwritten forms, and the facility issues physical membership cards while administrative staff manually log subscription dates into individual spreadsheets. This management model creates extreme operational friction when tracking, updating, or verifying expiration timelines as the lounge's active user base expands. Because there is no centralized database engine to link these operations, front-desk personnel must constantly context-switch between digital chat windows, physical logbooks, and desktop spreadsheet rows simply to execute a single check-in or validate a membership tier. This procedural friction renders daily front-office operations heavily time-consuming and highly susceptible to data entry errors.

As a result, the La Paz branch faces persistent operational liabilities that threaten its business performance and customer satisfaction metrics. These liabilities manifest as accidental room double-bookings, overlooked or missed client reservations, asynchronous data updates, and prolonged queues at the service counter. Managing overlapping transaction records across disconnected media significantly inflates the administrative workload of staff, leading to severe service bottlenecks during high-volume operating hours. These systemic inefficiencies collectively highlight an urgent, immediate requirement for a structured, unified, and permanent technological intervention.

To address these concerns, the proposed Integrated Study Hub Management System for WS Students & Professionals Lounge aims to centralize and automate key operations such as registration, reservations, membership management, billing, tracking time duration/expiration, and sales report generation. By replacing manual processes with a digital system, the lounge can significantly reduce errors, streamline workflows, and improve service efficiency. Ultimately, the system is designed to make operations easier for staff while providing a faster, more convenient experience for customers.

1.2 Background of the Study

Study hubs and co-working spaces have experienced an unprecedented surge in popularity. This growth is largely driven by the increasing demand from students, freelancers, and working professionals for structured, quiet, and highly productive environments that foster deep focus and collaboration. These spaces fulfill a critical societal and infrastructure need by providing essential facilities, including high-speed internet connectivity, ergonomically sound work areas, proper lighting, and specialized meeting rooms. However, as customer volume expands, the operational complexity of managing these shared facilities scales accordingly. To sustain high service quality and maintain customer satisfaction, modern study hubs are increasingly required to manage their day-to-day administrative frameworks with a high degree of operational efficiency.

WS Students & Professionals Lounge – La Paz Branch is one of the study hubs that offers space for studying, meetings, and collaboration. At present, however, the establishment relies almost exclusively on a combination of fragmented, manual procedures, handwritten logbooks, and legacy spreadsheet software to administer its daily operations. Furthermore, the lounge utilizes external social media channels—primarily Facebook and Facebook Messenger—as its primary interfaces for handling client inquiries and booking requests. This current operational framework spans across multiple administrative silos, including customer registration, workspace reservations, membership subscription tracking, elapsed time logging, invoicing, and retrospective financial reporting. Because these workflows are fundamentally decentralized and detached from one another, data retrieval and verification processes remain heavily disorganized and administratively cumbersome.

This reliance on non-integrated platforms exposes the establishment to significant operational vulnerabilities. Frontline personnel must constantly navigate and manually cross-reference communication threads on social media, physical logbooks, and individual digital spreadsheets simply to verify a single customer transaction or room availability status. This structural fragmentation frequently results in serious service bottlenecks, such as accidental resource double-bookings, delayed response times to online clients, prolonged front-desk waiting lines, and chronic inaccuracies in tracking real-time room usage and membership status. The logistical strain is particularly magnified during peak operating hours when the high influx of clients drastically increases the administrative workload, rendering the system acutely susceptible to human calculations and transcription errors.

To resolve these documented administrative bottlenecks, there exists a critical and immediate imperative to transition from a manual, paper-reliant architecture to a secure, digitized infrastructure. The proposed Integrated Study Hub Management System for the WS Students & Professionals Lounge is directly engineered to address these operational vulnerabilities. By consolidating and automating the core business mechanisms of client registration, interactive reservations, membership management, time-duration tracking, instant billing computation, and real-time sales reporting into a centralized web-based platform, the system establishes an integrated, database-driven workflow. Ultimately, this system is designed to improve operational efficiency, reduce errors, and enhance overall service quality for both staff and customers.

1.3 Statement of the Problem

As the demand for study hubs and co-working spaces continues to grow, businesses like WS Students & Professionals Lounge face significant operational challenges due to the reliance on manual and paper-based management systems. The current lack of a centralized platform for managing reservations, memberships, payments, and sales often leads to double bookings, long waiting times, and inaccuracies in tracking room usage and sales reports. Because most records are maintained in physical logs or scattered spreadsheets, the business is prone to data loss, human error, and delays in service delivery.

Without an integrated system, the staff must spend excessive time manually verifying records and updating information, which leads to operational inefficiency and customer dissatisfaction, especially during peak hours. Furthermore, the absence of real-time reporting makes it difficult for management to monitor sales and occupancy accurately, hindering their ability to make data-driven decisions. To maintain its reputation and provide quality service, it is essential for the lounge to transition into an automated and secure management framework.

Specifically, this study will seek to answer the following questions:

1. How can a real-time reservation system be designed and implemented to effectively prevent double bookings, optimize table allocation, and minimize the waiting time experienced by customers?

2. In what ways can a membership tracking system be developed to automatically monitor user activity, track usage duration, and manage subscription validity accurately?
3. What methods can be applied to integrate billing and sales reporting so that every transaction is automatically recorded and updated, minimizing manual errors?
4. What approaches can be employed to implement a centralized dashboard that displays real-time information on time usage, occupancy, and sales to assist management in decision-making?

5. How can a web-based management system improve the overall service quality and customer satisfaction of WS Students & Professionals Lounge compared to its current manual processes?

6. What technical features can be implemented to ensure that data and transaction records are securely stored and accessible only to authorized personnel?

7. In what ways does the system improve customer experience in terms of faster transactions, reliable booking, and efficient service delivery?

8. To what extent does the Integrated Study Hub Management System enhance the management of registrations and memberships?

9. What is the effectiveness of the system in handling billing, time-based room usage, and expiration tracking?

10. What impact does the system have on the accuracy and efficiency of sales report generation?

1.4 Objectives of the Study

1.4.1 General Objective

The primary objective of this study is to design and develop an Integrated Study Hub Management System for WS Students & Professionals Lounge – La Paz Branch. The proposed system aims to enhance operational efficiency by automating and streamlining core business processes, improve the quality and speed of service delivery, and ensure accurate, reliable, and centralized management of user registration, membership records, billing transactions, time-based room usage, and sales report generation. Ultimately, the system seeks to provide a more organized and efficient platform that supports both administrative functions and customer needs.

1.4.2 Specific Objectives

Specifically, this study will aim to:

- Design and develop a centralized web-based management system that serves as a single platform for all lounge operations and data storage.
- Automate the reservation and booking process to effectively eliminate double bookings and optimize table allocation for customers.

- Implement an automated membership and time usage tracking feature to monitor customer subscriptions and facility usage duration accurately.
- Develop an integrated billing and sales recording system that minimizes human errors and ensures that all financial transactions are logged correctly.
- Generate comprehensive real-time reports for sales, room occupancy, and time tracking to assist management in data-driven decision-making.

- Conduct a system of evaluation through user testing to ensure that the developed application meets the functional requirements and provides a user-friendly experience for the staff and administrators.

1.5 Scope and Limitations

This study focuses primarily on the design, development, and implementation of an Integrated Study Hub Management System engineered specifically to optimize the administrative framework and customer-facing workflows of the WS Students & Professionals Lounge – La Paz Branch. Designed as a centralized and the project directly addresses the systemic operational vulnerabilities inherent in manual, legacy business setups.

The system aims to automate traditionally labor-intensive workflows such as handwritten reservation books, manual elapsed-time calculations, and paper-based invoicing thereby drastically minimizing human mathematical errors, mitigating data fragmentation, and eliminating accidental resource double-bookings. By modernizing these core daily transactions into real-time digital processes, the developed platform enhances overall data accuracy, secures historical transaction records, and optimizes resource allocation. Ultimately, this integration elevates operational efficiency for frontline staff while providing management with the transparent, data-driven oversight necessary to maintain superior service quality and business integrity. The scope of the system will include the following features:

1. **Reservation Management** – This feature delivers an interactive, functional module allowing both frontend lounge staff and registered clients to schedule and secure specific study spaces, study pods, or conference rooms in advance. It incorporates programmatic validation logic to prevent double-bookings, handle reservation overlaps, and optimize physical table allocation in real time.

2. **Membership and Time Tracking** – This module securely records detailed customer profiles and automatically monitors physical usage durations via server-side check-in and check-out timestamps. It programmatically tracks subscription validity periods and membership tier benefits to ensure that workspace privileges are validated automatically before a session begins.

3. **Payment Billing** – This component handles the internal financial workflow of the lounge. It dynamically converts the accumulated duration from the time logs into itemized financial charges based on the user's applicable hourly rate or membership status, ensuring that all individual customer transactions are calculated and recorded accurately.

4. **Sales Recording and Reporting** The system permanently logs all financial transactions to systematically compile analytical summaries. It generates real-time, tamper-proof sales reports categorized by daily, weekly, or monthly intervals, providing management with accurate ledger data to monitor business performance effectively.

5. **Admin and Staff Dashboard** - The system features distinct centralized interfaces tailored to user roles. The control screens display live metrics regarding room occupancy, active workspace countdowns, real-time sales summaries, and automated time-expiration alerts to facilitate seamless front-desk operations and high-level administrative decision-making.

6. **Role-Based Access Control (RBAC)** – To reinforce system security and data integrity, strict access permissions are programmatically enforced based on user type. System administrators are granted full access to core configurations, analytical reports, and inventory parameters, while frontline staff members are restricted to purely operational features such as logging walk-ins, executing bookings, and processing billings.

1.5.1 Limitations may include:

1. **Geographic Limitation** – The system is exclusively designed and configured around the business model and localized workflows of the WS Students & Professionals Lounge – La Paz Branch only. The operational rules, spatial layout, and membership rules are bound to this single branch and do not support multi-branch scaling, cross-branch data syncing, or shared roaming memberships.

2. **Platform Limitation** – The application is strictly developed as a web-based system. While the user interfaces are designed responsively to properly render on mobile screens (smartphones and tablets), the scope completely excludes the development of native mobile applications published on the Apple App Store (iOS) or Google Play Store (Android).

3. **Internet Dependency** – Because the application relies on a cloud-hosted framework and a centralized relational database to prevent fragmented records, the system exhibits absolute internet dependency. The system requires a stable internet connection to process transactions; it does not feature an offline-first synchronization mode or local peer-to-peer data backup capabilities.

4. **No Accounting Integration** – The software focuses strictly on day-to-day operational logistics (reservations, memberships, time tracking, basic billing, and historical sales logs). It does not function as an enterprise resource planning (ERP) system and completely excludes advanced accounting integrations, such as balance sheets, profit-and-loss statements, tax computations, or government payroll deductions.

5. **No Hardware Automation:** The platform operates purely at the software application layer. It does not integrate with physical internet-of-things (IoT) devices, biometric fingerprint sensors, RFID card scanners at the entrance, or automated electronic door strikes for the study rooms. Physical access verification and room tracking remain under the manual oversight of front-desk personnel.

1.6 Significance of the Study

The findings of this study will bring significant improvements to the operational workflow of WS Students & Professionals Lounge – La Paz Branch. By transitioning from a manual, paper-based setup to an Integrated Study Hub Management System, the organization will benefit from a more organized and centralized platform. This transition is expected to address the recurring challenges of double bookings and long waiting times, which will ultimately allow the lounge to deliver faster, more efficient, and more reliable services to its clients.

The implementation of this system will provide a secure and efficient way to manage membership data and financial transactions. It will help the management and administrators gain stronger control over their daily operations through a real-time dashboard that displays sales, occupancy, time-usage information, and daily analytic reports. This transparency will help build trust with customers and will ensure that the business can maintain accurate records without the risk of manual errors or data loss.

Furthermore, this study will contribute to the field of Information Technology by demonstrating how automation can solve real-world business problems. It will serve as a practical reference for Future Developers and Researchers who may face similar challenges in developing management systems for the service industry. By providing improved features for reservation tracking and automated billing, this project will highlight the importance of digital transformation in enhancing customer satisfaction and operational efficiency.

1.7 Definition of Terms

1. Administrator – The management or authorized corporate personnel of WS Students & Professionals Lounge who possess the highest level of system configuration privileges. In this platform, the role is responsible for overseeing the business health of the La Paz branch by managing user access credentials (RBAC), generating consolidated sales reports, and auditing historical system logs to ensure operational accountability.

2. Analytical Reports – The systematically generated data summaries, data visualizations, and tabular charts within the system that translate raw operational logs into actionable business insights. In this study, these reports compile and analyze aggregated system metrics to visually map out lounge space usage trends, peak operating hours, and customer demographic metrics (such as the ratio of students to working professionals) for administrative decision-making.

3. Centralized System – A computing architecture where all data processing, storage, and management logic are concentrated within a single database and server infrastructure. In this study, it refers to integrating the separate functions of the WS Lounge—specifically room reservations, time tracking, inventory control, and billing—into one unified platform, ensuring that updates made by users or staff are instantly reflected across all dashboards.

4. Database – The relational storage architecture developed using MySQL where all structured information critical to the lounge's operations is systematically stored and

protected. This includes the secure preservation and real-time retrieval of interconnected tables concerning lounge members' profiles, physical check-in and check-out timestamps, historical financial transactions, and authorized staff accounts.

5. Membership Plans – A dedicated functional module within the system that actively monitors and manages the subscription lifecycles, user profiles, and associated tier benefits of the lounge's long-term clients. It automatically updates subscription validity windows, tracks recurring billing cycles, and validates active statuses during the check-in process to ensure members instantly receive their assigned workspace privileges and discounts.

6. Operational Bottlenecks - The specific workflow delays, structural inefficiencies, and administrative errors experienced in the manual, legacy setup of the lounge. In this study, it refers directly to resource conflicts like accidental double-bookings of study rooms, human mathematical errors during manual checkout time computations, and slow, paper-based invoice generation that prolongs customer wait times at the front desk.

7. Reservation System - The interactive, functional component of the web platform that permits both frontline staff and general clients to securely book specific study pods or conference rooms in advance. It features real-time availability calendars and automated overlap validation logic to ensure that selected timeslots are locked immediately upon approval, eliminating scheduling conflicts..

8. Role-Based Access Control (RBAC) A built-in system security feature that restricts data access and modification privileges based on a defined user hierarchy. It programmatically separates the dashboard interfaces, screen views, and functional capabilities among three distinct roles: the Administrator (full system configuration,

revenue audits, and inventory management), the Front-desk Staff (check-ins, check-outs, and manual order entry), and the General Customers (viewing availability, booking reservations, and tracking personal monthly membership status)..

9. Sales Report – The auto-generated financial audit summary within the system that programmatically compiles and aggregates revenue earned over daily, weekly, or monthly intervals. It consolidates individual income streams generated from workspace room rentals, monthly membership subscriptions to provide the Administrator with a transparent ledger for business analytics and financial tracking.

10. Study Hub Management System - The specific, web-based software platform developed for the WS Students & Professionals Lounge (La Paz Branch) to replace manual administrative workflows. It centralizes and automates five core operational subsystems: user registrations (account profiling), space reservations (conflict-free room bookings), monthly membership subscriptions, time-based logging (automated check-ins and duration calculations), and consolidated billing.

10. **Time Logs -** The automated database records that programmatically capture and store the exact check-in and check-out timestamps of clients. In this study, these logs are used by the system's core billing engine to dynamically calculate precise workspace usage

durations and convert that time into accurate, itemized financial charges based on the user's applicable hourly rate or membership tier.

11. **User Dashboard -** The personalized, user-facing control screen accessible exclusively by authenticated lounge clients upon logging into the platform. This interface serves as a secure self-service portal, aggregating real-time data from the MySQL database to allow students and professionals to independently view their active room reservations, track remaining membership validity days, and browse their itemized past billing and transaction histories.
12. **Web-Based System -** The architectural deployment framework of the developed application, which allows the lounge management platform to be securely accessed over an internet connection using standard web browsers. This setup removes the need for local software installations, enabling administrators, staff, and clients to use the system across multiple cross-platform devices, including laptops, tablets, and smartphones.

CHAPTER 2

Review of Related Literature and Studies

Overview of an Integrated Study Hub Management System for WS Students & Professionals Lounge

A study hub management system represents a specialized class of administrative software designed to orchestrate, automate, and centralize the diverse operational workflows within modern co-working and shared-facility workspaces (Sommerville,

2016). In contemporary business and academic environments, these systems have evolved from simple scheduling applications into comprehensive platform ecosystems. They allow business owners, frontline personnel, and clients to seamlessly coordinate resource allocations, handle memberships, and process financial exchanges regardless of location.

To cater to varying business demands, workspace management systems utilize diverse digital infrastructures:

- **Cloud-Hosted Booking Infrastructures:** A third-party framework that provides rapid, remote access to real-time space availability, allowing clients to secure reservations dynamically via any internet-enabled device (Laudon & Laudon, 2021).
- **Cross-Platform Mobile Integration:** A method optimized for mobile devices, enabling clients to track their remaining membership duration, receive push notifications, and monitor current check-in statuses on the go.

- **Administrative Desktop Command Centers:** A stationary control platform engineered for frontline lounge staff, utilized to handle sudden walk-in clients, audit facility check-outs, and monitor general utility usage within the hub.
- **Local Relational Databases:** A structured data architecture that serves as the backbone of the lounge, securely holding financial logs, user authentication files, and active time-tracking sheets to ensure high data consistency (Silberschatz et al., 2019).

In the context of daily business operations within co-working facilities, relying on manual transaction frameworks creates significant operational bottlenecks. Paper-based logbooks and fragmented communication channels (such as manual social media message logging) are heavily susceptible to human oversight errors. These errors frequently manifest as overlapping space reservations (double bookings), inaccurate manual

timekeeping, disorganized customer profile tracking, and slow invoice generation (Heizer et al., 2020). For example, when multiple clients attempt to secure a group conference room simultaneously without a real-time ledger, it often leads to operational conflicts, scheduling delays, and an eventual decline in customer trust.

The strategic deployment of an integrated, automated management system completely mitigates these administrative risks by enforcing precise digital validation rules. By automating the lifespan of a booking—from the initial online reservation to the final algorithmic billing calculations upon check-out—the system guarantees structural data integrity and transaction transparency (Romney & Steinbart, 2020). Digital platforms drastically accelerate transaction speeds, lower the business's dependence on physically tracking paper logs, and eliminate risks concerning misplaced receipts or corrupted customer records. By resolving these operational vulnerabilities, the management system protects fragile enterprise financial data, enhances staff productivity, and guarantees a seamless, modern, and reliable environment for both student and professional clients.

2.1 Foreign Studies

Management Information Systems: Managing the Digital Firm

The study by Kenneth C. Laudon and Jane P. Laudon (2021) provides a foundational understanding of management information systems (MIS) and their role in modern organizations. In their book, Management Information Systems: Managing the Digital Firm, the authors explain that information systems are essential tools that integrate people, processes, and technology to support organizational operations and decision-making. A digital firm is one where nearly all significant business relationships with customers, suppliers, and employees are digitally enabled and mediated. True success comes from realizing that technology is an enabler, but the organizational design and management strategy dictate the actual return on investment. This concept is highly relevant to the development of an integrated study hub management system, as such a system requires efficient coordination of users, resources, and services.

Laudon and Laudon further emphasize that digital systems improve operational efficiency by automating routine tasks and centralizing data management. In the context of a study hub or lounge for working students and professionals, an integrated system can streamline processes such as reservations, attendance tracking, resource allocation, and user management. By reducing manual work and minimizing errors, the system enhances overall service quality and user experience.

Moreover, the authors highlight the importance of data-driven decision-making. Information systems enable organizations to collect and analyze data, allowing administrators to monitor usage patterns, peak hours, and user preferences. This capability is essential for managing a study hub effectively, as it helps optimize space utilization, improve services, and support strategic planning.

Additionally, Laudon and Laudon discuss how emerging technologies contribute to innovation and competitive advantage. Implementing an integrated management system for a study hub aligns with this idea by leveraging digital solutions to meet the needs of modern users, particularly working students and professionals who require flexible, efficient, and technology-supported environments.

Overall, the concepts presented by Laudon and Laudon (2021) support the need for developing an integrated study hub management system, as such systems enhance operational efficiency, improve user satisfaction, and enable data-driven management.

Reference: Kenneth C. Laudon, Jane P. Laudon, Management Information Systems, 15 Ed., © 2018, Pearson Education, Inc., New York, NY. ISBN 978-0-13-463971-0.

Information Technology for Management Driving Digital Transformation to Increase Local and Global Performance, Growth and Sustainability

Information Technology for Management is an excellent, comprehensive guide on how modern enterprises bridge the gap between core business strategy and emerging tech infrastructure. The study of Turban, E., Pollard, C., & Wood, G. (2021). Digital Transformation Disrupts Companies, Competition, and Careers Locally and Globally. Information Technology for Management discusses the importance of aligning business-IT strategies and explains how companies rely on data, digital technology, and mobile devices to support them in the on-demand and sharing economies. Their goal is to provide students from any business discipline with a strong foundation for understanding digital technology concepts and terminology and the critical role that it plays in facilitating business sustainability, profitability, and growth locally and globally. Many forward-thinking managers and entrepreneurs are digitally transforming their existing business models and reinventing their businesses. By no longer operating and maintaining outdated and complex IT architectures with a mix of legacy systems that can delay or prevent the release of innovative new products and services and absorb large portions of the information technology (IT) budget, companies can add value, increase their customer base, expand their business capabilities, and increase profits.

As businesses continue to change their business models to accommodate the needs of the on-demand and sharing economies, IT professionals must constantly scan for innovative new technologies to provide business value, help shape the future of the business, and facilitate performance and growth in local and global markets. For example, smart devices, mobile apps, sensors, and technology platforms along with increased customer demand for digital interactions and on-demand and shared services have moved commerce in fresh new directions.

The differences between the on-demand and sharing economies and the six business objectives IT should focus on to enhance organizational performance, growth, and sustainability. The on-demand economy is the economic activity created by technology companies that fulfill individual consumer demands through the immediate provisioning of products and services. In the sharing economy, goods or services are shared between private individuals through an online company or organization. The six business objectives that IT should focus on are as follows:

1. Product development to help businesses respond quickly to changing customer demands.
2. Stakeholder integration to communicate with shareholders, research analysts, and others in the market.
3. Process improvement to increase efficiency and cost effectiveness of internal business processes.
4. Cost efficiencies to reduce transaction and implementation costs.
5. Competitive advantage to bring a product to market cost effectively and quickly.
6. Globalization to stay in contact with its global employees, customers, and suppliers 24/7.

The role of IT in improving business processes. Understand the concepts of business process reengineering and competitive advantage. Outdated and complex application architectures, with a mix of interfaces, can delay or prevent the release of new products and services, and maintaining these obsolete systems absorbs large portions of the IT budget. As a result, managers and entrepreneurs must integrate digital disruptive technology into their products and services to improve their business processes and stay competitive BPR is the concept of using IT to radically improve processes rather than simply making incremental positive changes. Competitive advantage is when an organization differentiates itself by charging less and creating and delivering better quality products or services than its competitors.

Reference: Turban, E., Pollard, C., & Wood, G. (2021). Information technology for management: Driving digital transformation to increase local and global performance, growth and sustainability (12th ed.). John Wiley & Sons. ISBN-13 **978-1119802525**.

Developing and Implementing Web-based Online University Facilities Reservation System

The study by Alkhaldi et al (2018). Community services have recently grown through government and private institutions due to an increased awareness of their importance in societal development. These distinctive services are now available free of charge, expanding beyond financial assistance to encompass various other forms of community support. Ultimately, these diverse services are vital for the progress and prosperity of society by helping individuals become active and constructive citizens.

Recognizing the vital importance of community engagement, universities actively seek to provide diverse social services to their surrounding populations. These initiatives include establishing free accredited courses for students and hosting essential secondary school entrance evaluations like the General Aptitude Test (GAT) and Qiyas Tests. To further spread this community service culture, Imam Abdulrahman Bin Faisal University (IAU) has proposed a system allowing employees and visitors to reserve various campus facilities, including halls, courts, swimming pools, and theaters. Through this centralized platform, users can seamlessly book facilities, cancel reservations, track requests, and manage their accounts while the university ensures all venues are continuously monitored and maintained.

The main motivation of the work is that there is no online reservation system for all IAU facilities, through the proposed system; the customer can reserve the IAU facility from any place and at any time according to facilities schedule, this will increase the utilization of university facilities by the community members. Thus, an online facilities reservation system for IAU is necessary. The facilities reservation system should provide user friendly interface and complete functions.

The IAU system should be able to satisfy the requirement in providing the suitable functionality to the user, where user should be able to know the availability schedule of different IAU facilities, user should be able to reserve a facility needed online anywhere and anytime, and user should be able to check his/her reservation schedule online.

Alkhaldi et al (2018). The work presents the design and development of the online Imam Abdulrahman Bin Faisal University (IAU) facilities reservation system. The proposed platform successfully resolves the imbalanced utilization of campus venues like halls, theaters, swimming pools, and stadiums. Built using Unified Modeling Language (UML), MySQL, and the Visual Basic (VB) programming language, the system addresses the university's prior lack of centralized facility management. It allows local community members to easily reserve these campus spaces free of charge, fulfilling the university's non-profit mission of public service. Furthermore, this system framework can be extended to other departments within IAU, or adapted by external universities and companies to optimize their own resource management.

Reference: Alkhaldi, D., Alkhaldi, Dhai., Aldossary, H., Alsmadi, M., Al-Marashdeh, I., Badawi U, A., Alshabanah, A., and Alrajhi, A. (2018). International Journal of Applied Engineering Research ISSN 0973-4562 Volume 13, Number 9 (2018) pp. 6700 6708.

RESTAURANT MANAGEMENT SYSTEM FOR KAS VALLEY RESTAURANT

The recent advancement in technology and its tremendous impact, many businesses have integrated technological tools into their business operations. The food industry is no exception. In the global food industry, technology has made it possible to improve food services, increase customer satisfaction and maximize business returns. According to Alormene C.K (2022). Build resilient infrastructure, promote sustainable industrialization and foster innovation.

In today's fast-paced economy, technology has become a central element for optimizing business operations and maximizing customer satisfaction across various industries. Within the food sector, digitalization has enabled many restaurants to launch online ordering services via dedicated websites and applications. The critical need for these digital systems became undeniably evident during the COVID-19 pandemic when temporary shutdowns starved businesses of in-person customers. Lacking alternative digital infrastructure to support their operations, several traditional restaurants unfortunately went out of business.

Kas Valley Restaurant is struggling to efficiently manage its business operations as growing consumer demand shifts toward faster, technology-driven transactions. The restaurant currently relies on manual channels like walk-ins, phone calls, and social media to process all incoming customer orders. However, these methods have caused excessively long physical queues and frustrating phone delays due to poor network reception or busy lines. Furthermore, Kas Valley lacks a centralized record-keeping platform, making it difficult to track order data or generate accurate sales analytics. Managing table reservations is also an arduous process because staff must manually check records to determine booking availability. Ultimately, these combined administrative inefficiencies have significantly slowed down daily operations and decreased overall customer satisfaction.

While advancements in restaurant software and POS technologies have improved global customer service, the Ghanaian food industry has not fully utilized these emerging technologies to optimize operations. The few Ghanaian restaurants that do use software typically limit their applications to basic POS transactions or isolated online ordering systems. Consequently, these applications leave out core operational needs such as table reservations, customer data management, and e-commerce integration. This lack of a centralized system creates major issues with data inconsistency, data redundancy, and scattered information across different units. Ultimately, these fragmented systems make business data analysis difficult for management, which harms the restaurant's overall efficiency and profitability in the long run.The project aims to develop a web-based restaurant management system that facilitates online and in-person meal ordering in Kas Valley restaurant. The application will act as a centralized system for maintaining their records, including customer data. The system also aims to support table reservation management and supply actionable business insights through data analytics on the business s sales.

Reference: Alormene C.K (2022). RESAURANT MANAGEMENT SYSTEM FOR KAS VALLEY RESTAURANT. Department of Computer Science, Ashesi University.

Design and Implementation of Reservation Management System Case Study: Grand Ville Hotels

The advances in mobile technology and availability of broadband have made it easier to search for a hotel and book through an online reservation system. Although mobile applications are unlikely to revolutionize a hotels' reservation system to the same extent that the internet did, it may have an impact on hotel revenue management and the concept of implementing price structures. The study of Wiilams, K and Ajinaja, M (2019) The management and booking of Rooms in hotels is a tedious and complicated task especially if it is done manually. Keeping track of large customers and all their details requires an inordinate space for file cabinets, not to mention the time the hotel administrator would spend going back and forth to file cabinets so as to look up each customer's information. This is why a good hotel reservation system is needed to make this task as easy as possible. The system was developed using the object-oriented software development approach which includes the use of object-oriented analysis, object- oriented design and object oriented programming. This was done so that the developed software can be maintainable, reliable and scalable.

Wiilams, K and Ajinaja, M (2019) An effective reservation system has to employ the latest technology to allow hoteliers to gain a competitive advantage over their rivals and remain at the forefront of emerging trends. Hotels must therefore adapt their websites to respond to this development by adopting the use of booking engines and developing marketing strategies to include mobile technology. This will help hoteliers to control all reservations that come from different sources in a single control system, distribute rooms favorably to different online channels and analyze the contribution each sales channel provides to revenue, identifying the most efficient ones.

Room reservation is the most critical component of a hotel system, requiring a dynamic and efficient process to maximize overall profitability. The front office manager is responsible for overseeing this system, which primarily handles two distinct customer types: individual guests and group bookings. During peak seasons, stable group customers are given higher priority over individuals to ensure more secure and predictable revenue. To navigate these demands, the manager relies heavily on a computerized system that provides the necessary data and tools to make optimal room allocation decisions. Ultimately, these digital tools empower hotel leadership to streamline the reservation process and successfully meet their commercial goals.

Grand villa Hotels which is the purported case study used improvedmanual system in their room booking. During interview session with the manager of the hotel, some of the problems that were identified by using the Excel sheet includes:

- Lack of immediate retrievals: Since different excel sheets were used to store different transactions e.g. room reservation, food ordering, customer details, it is often hard to retrieve and to find particular information quickly. For example, to find out about the customer's room booking details, the user has to go through various excel sheets. This results in inconvenience and wastage of time.
- Lack of immediate information storage: The information generated by various transactions takes time and efforts to be stored at the right place.
- Lack of prompt updating: Various changes to information like customer's details or cancellation of reservation are difficult to make.
- Preparation of accurate and prompt reports: It takes a lot of time to compile information.

Hotel needs to maintain the record of guests and reserve rooms beforehand. Customers should be able to know the availability of the rooms on a particular date. They should be able to reserve the available rooms according to their need in advance so as to make their stay comfortable. No hotel reservation system can operate alone in the hotel, because it concentrates mainly in reservation process, the system needs an accounting system and a management information system, and so it can serve all the needs of the hotel. It is therefore very important to build new and modern flexible dynamic effective compatible reusable information systems including database to help manipulate different processes and operations that are carried out in hotels.

Reference: Wiilams, K, and Ajinaja, M. (2019). Design and Implementation of Reservation Management System Case Study: Grand Ville Hotels. Journal of Information Engineering and Applications. ISSN 2225-0506.

Design and Implementation of a Reservation System and a New Queuing for Remote Labs

According to Khazri, Y., Fahli, A., Moussetad, M., and Naddami, A. (2019). The fast development of the Internet technology and its increase of popularity has an enormous impact on the process of teaching.

Modern technology offers innovative tools that enhance educational strategies, with internet and mobile technologies fostering practical scientific and technical training through interactive applications. Consequently, e-learning has become an important component of practical education, offering remote laboratories as a viable alternative to traditional face-to-face training. These distributed systems help solve the challenges of student massification by allowing institutions to share human and material resources while providing real-time online access to practical work.

These computing and materials-based environments complement conventional training by enabling students to manipulate real, dangerous, or expensive devices remotely. Their development facilitates flexible, hands-on learning for a growing student population despite limited material and coaching resources. Additionally, they allow institutions to share heavy equipment nationally or internationally while expanding available presentation methods.

Remote laboratories function as computing and materials-based environments that complement conventional training by enabling students to manipulate real, dangerous, or expensive devices from a distance. The deployment of these facilities accommodates a growing student population by facilitating flexible, hands-on learning despite shortages in material and coaching resources. Furthermore, this approach allows institutions to share heavy equipment on a national or international scale while simultaneously expanding available presentation methods.

Virtual and remote laboratories reduce the costs associated with conventional hands-on facilities by minimizing the need for expensive equipment, space, and maintenance staff. These platforms also enhance engineering and scientific education by supporting distance learning, improving accessibility for handicapped individuals, and ensuring safety during dangerous experiments. To address these needs, the authors implemented the ELAB FSBM platform, which features a user queue management system and enables collaborative, supervised remote practical work for both teachers and learners.

Khazri, Y., Fahli, A., Moussetad, M., and Naddami, A. (2019). Design and Implementation of a Reservation System and a New Queuing for Remote Labs. iJOE – Vol. 15, No. 12. https://doi.org/10.3991/ijoe.v15i12.11098.

PROJECT MANAGEMENT IN PROTOTYPING AN ONLINE TABLE RESERVATION SYSTEM

The rapid digitalization of the food and beverage industry has transformed operational management and customer interactions through smart, data-driven platforms for ordering, payment, and customer relations. Felix, A., Leornardo, V., Melcelyn., and Ananda, D. (2026). Despite this ongoing digitalization, many restaurants still rely on manual methods to manage table reservations, such as phone calls, walk-in requests, or unstructured messaging applications. These manual processes often lead to miscommunication, double bookings, long queues, and difficulties in predicting demand during peak hours. As a result, customers may experience long waiting times and dissatisfaction, while restaurant owners struggle to optimize seating capacity, staff allocation, and raw material planning.

Previous studies have examined online reservation systems and digital solutions for restaurant management, including web based booking platforms, smart service technologies, and integrated restaurant management applications. These works generally focus on the design and development of reservation systems, the use of specific technologies, or the impact of digitalization on service quality and business performance. However, most of them discuss technological and managerial aspects separately and do not explicitly adopt a comprehensive project management framework to guide the development and implementation of online table reservation systems.

This situation highlights a clear research gap in how project management standards, like PMBOK, are systematically applied to developing online table reservation systems within the food and beverage sector. Consequently, there is a strong need for studies that combine both technical system development and structured managerial frameworks to address scope, time, cost, quality, risk, and stakeholders simultaneously. These findings are consistent with Akyüz and Balkan (2024), who emphasize that smart service technologies in hospitality require not only robust technical design but also structured project management to achieve operational efficiency.

Felix, A., Leornardo, V., Melcelyn., and Ananda, D. (2026). This study developed a web based online table reservation system for the food and beverage (F&B) industry by integrating the Project Management Body of Knowledge (PMBOK) framework with the Systems Development Life Cycle (SDLC). The main findings show that the proposed system can reduce miscommunication in reservations, shorten waiting times, and provide more transparent information about table availability and restaurant details. The evaluation results indicate that users perceive the system as easy to use, responsive, and more reliable than manual phone or walk-in reservation methods.

This research theoretically contributes to the literature on hospitality and F&B digitalization by showing how PMBOK project management standards can be applied to online reservation systems. It demonstrates that core knowledge areas—such as Scope, Time, Cost, Quality, and Risk Management—can be directly mapped onto concrete stages of system design. By doing so, the study successfully bridges the long-standing gap between purely technological and managerial perspectives. Ultimately, this integrated view enriches existing literature, which frequently discusses digital transformation and system features at a much more fragmented level.

Practically, these findings offer valuable guidance for F&B businesses and system developers planning to implement online table reservation solutions. The included system design, evaluation insights, and project documentation (such as the WBS, Gantt Chart, and budget planning) can serve as a functional reference model for similar efficiency-focused projects. Ultimately, adopting this structured project management approach allows restaurant managers to optimize resources, mitigate implementation risks, and ensure the new digital system delivers tangible business benefits.

Felix, A., Leornardo, V., Melcelyn., and Ananda, D. (2026). PROJECT MANAGEMENT IN PROTOTYPING AN ONLINE TABLE RESERVATION SYSTEM. Indonesian Interdisciplinary Journal of Sharia Economics (IIJSE). e-ISSN:2621-606X. Vol. 9. No. 2(2026). Page: 13325-13347.

Online Reservation Management System to Increase Transaction Efficiency at MOMENKITA Photo Studio

In an increasingly competitive and complex business environment, companies around the world are looking for innovative ways to improve their operational efficiency. The key to achieving sustainable growth and maintaining competitiveness in the global market lies in corporate efficiency. The study by Firmansyah et al. (2024). "Momenkita Studio" is a small and medium-sized enterprise (SME) operating in the creative industry, specifically as a photo studio. It faces challenges with management that has not yet been optimized. Some noticeable obstacles include the lack of a structured customer record system, often complicated scheduling, and unclear monitoring of studio activities

One increasingly relevant solution to achieve this goal is through the implementation of Web-based corporate management applications, especially in photo studio companies. This inefficient management can lead to schedule overlaps, neglect in record-keeping, and difficulties in tracking payment transactions. This study uses a qualitative method. Therefore, there is a need for a solution that can improve management efficiency.

"Momenkita Studio" in scheduling photo sessions and meetings with customers is still done manually, so it becomes complicated without an effective schedule management tool. Presentation and portfolio management are difficult in physical form, and the studio must rely on photo prints. So website-based company management is needed to overcome existing problems. This application is designed to help companies overcome challenges in customer management in an efficient and structured way. This application allows for more efficient collection and storage of customer data.

Firmansyah et al. (2024).The focus of the research is on the design and implementation of company management applications which include the design of customer management, namely the photo shoot schedule reservation system, which is done online, transaction management, namely payment for photo shoot services, purchase of studio equipment, report management, namely customer reports, weekly, monthly, and annual transaction reports, user management, namely user access rights for the application. In addition, this research will provide a deep understanding for researchers, business practitioners, and decision makers about how technology can be an effective medium in increasing company efficiency and strengthening customer relationships.

Firmansyah et al. (2024). This research developed a dedicated photography website that visually displays reservation schedule availability using an interactive, integrated calendar. An interview with the owner of MOMENKITA Foto Studio revealed that the platform aims to enhance service accessibility, simplify the booking process, and provide a more comfortable customer experience. To achieve these goals, the website features specialized tools including package selection, secure online payment integration, a portfolio photo gallery, clear service information, and easily accessible contact details.

Firmansyah et al. (2024). This research is able to provide solutions to the problems that exist in MOMENKITA Photo Studio, namely it can facilitate the need for information on the availability of reservation schedules because the schedule is visualized in the form of a calendar, reservation data reports that are automatically summarized so that they are better organized. The results of the study show that the concept of schedule visualization can make it easier for customers to get information so that they can minimize reservations with dates or times that are not available. Providing information via a website can provide more complete information and present it as more interesting information and can be used to help as a promotional medium.

Reference: Firmansyah, F., Razak, M,R,R., and Sofyan, W. (2024). Online Reservation Management System to Increase Transaction Efficiency at MOMENKITA Photo Studio. Journal of Accounting and Finance Management (JAFM). E-ISSN: 2721-3013. Vol. 5.

Online reservation systems in e-Business: Analyzing decision making in e-Tourism

The study by Halkiopoulos et al.(2020). Focuses on exploration of knowledge for online booking systems and on the views of local students-users concerning the booking rate based on these online systems. Another perspective of this project is to investigate the decision-making process (emotion-focused) that they follow in order to choose a tourist destination via online booking systems

The electronic commerce is defined as the activity of sale and marketing for products and services through an electronic system such as, for example, the Internet. It involves the electronic data transfer, the distribution management, e-marketing (online marketing), online transactions, electronic data changes, the automated inventory of used management systems, and automated data collection. E-Tourism (electronic tourism) is a part of electronic commerce and unites one of the fastest development technologies, such as the telecommunications and information technology, hospitality industry and the management / marketing / strategic planning.

This research aims to analyze students' perspectives on electronic booking tools and investigate their specific decision-making processes when planning travel through these digital platforms. While the continuous growth of online booking systems benefits both tourists and hotel owners alike, numerous studies have already been conducted to identify and evaluate how well these modern systems satisfy travelers' needs.

As a result ,this demonstrates how the expansion of the internet has significantly transformed market conditions by providing tourist organizations with innovative new tools for marketing and management. While this digital evolution facilitates direct interaction with users and changes the entire tourism development process, a review of existing literature reveals that the field of e-Tourism still faces numerous unanswered questions.

Knowing the cause and the consequence is, in simple terms, what should be achieved. In this direction many issues can and should be raised that question the relationship of the customers, their characteristics, the way they 'came' to the hotel (information search, booking and paying for accommodation) as well as various factors that led to their decision.

References: Halkiopoulos, C., Antonopoulou, H., Papadopoulos, D., Giannoukou, I. & Gkintoni, E. (2020). Online reservation systems in e-Business: Analyzing decision making in eTourism. Journal of Tourism, Heritage & Services Marketing, ISSN2529-1947, Vol. 6, No. 1, pp. 9-16. http://dx.doi.org/10.5281/zenodo.3603312.

The Dentist Online Reservation System Design and Implementation Web Based Application and Database Management System Project

Technology has changed many aspects of life and people's daily life is becoming indivisible from the network due to the development of Internet. With online dentist reservation system, the process gets much faster and more efficient than traditional way. Thousands of business and organizations have already discovered the advantage gained by using the online reservation system. The study by Kerdvibulvech, C., and Win, N,N. (2012). Online dentist reservation system allows the system administer to access and manage the database online, quickly pull data and create strong reports right from the online reservation system with most practical to find the fastest information, instead of having to maintain and manage separate data files, folders and spreadsheets. They simply navigate to the system just as any Web Site. The data will be housed securely and safely online.

The system also helps the administrator to maintain the database online easily. It enables the administrator to check the patient's requests, manage the appointment schedule, and manage the patient's information. It offers an opportunity for patients to submit their personal comments to the dental clinic to give better services. The design for the proposed system can be obtained easy way to access the enquire information about the dentist appointment.

Kerdvibulvech, C., and Win, N,N. (2012). The Online Dentist Reservation System was successfully built using Microsoft Access, ASP.NET 2010, SQL Server, and the C# programming language. Due to restricted financial resources and a tight timeframe, the developer could not fully implement all intended system functionalities. However, the system's flexible architecture allows junior students to pursue future work on the project and add new modules without disrupting the current setup. Potential enhancements include linking the booking and appointment tables to help dentists manage their schedules more easily. Finally, the system can be expanded to support online billing and allow patients to cancel, pre-pone, or post-pone their appointments directly through the platform.

Kerdvibulvech, C., and Win, N,N. (2012). The Dentist Online Reservation System Design and Implementation Web Based Application and Database Management System Project. International Conference on Education Technology and Computer (ICETC2012) IPCSIT vol.43 (2012) © (2012) IACSIT Press, Singapore.

Web Site &Web Based Hotel Reservation Management System For Sky Lodge Hotel

Accordin to P. B. N. N. De Silva (2017). Relaxation they say is the best medicine for the modern life style, which sharpens the quality of the lifestyle of an individual.

Most of the people use to travel and spend time by living with the beauty of the nature. Therefore, the hospitality providers have taken measures to improve the best available solutions for this purpose.

Tourism is a fast-growing industry in Sri Lanka and stands as one of the country's primary financial contributors. Recognizing the need for an organized institutional framework, the Government decided to develop the tourism sector in a planned and systematic manner in 1966. This initiative led to the official establishment of both the Ceylon Tourist Board and the Ceylon Hotels Corporation. The Ceylon Tourist Board operated as a statutory body designed to allow greater flexibility in financial management and organizational decision-making. Meanwhile, the Ceylon Hotels Corporation was formed as a joint stock company to serve as the government's commercial arm for expanding tourist accommodations and facilities.

The Sky Lodge is a prominent hospitality provider that caters to both local and foreign travelers. It is situated in the beautiful city of Gampola, which is located within the ancient cultural district of Kandy in Sri Lanka's Central Province. Guests staying at the hotel enjoy close access to highly sacred local temples, including the Temple of the Tooth Relic and the Bahirawakanda Temple. The surrounding region also offers proximity to scenic and historic attractions, such as breathtaking waterfalls, the Peradeniya botanical garden, and the ancient Bogoda Covered Bridge. Ultimately, the combination of strong cultural sights, lush mountains, diverse wildlife, and friendly locals makes the Sky Lodge an ideal destination for an unforgettable holiday.

P. B. N. N. De Silva (2017). The Sky Lodge Hotel's innovative concept led to an increased workload, which reduced task accuracy and created issues like booking errors and faulty payment calculations. Employees were forced to spend excessive time completing manual room reservations, leaving data vulnerable to damage and lowering overall guest satisfaction. To overcome these operational hurdles, a highly anticipated Web

Based Hotel Management System was introduced to replace the manual process. This proposed platform features centralized modules designed to efficiently manage hotel rooms, booking processes, employee registrations and leaves, hotel expenses, and administrative system activities.

After considering the complications and drawbacks of their existing manual system, the management of the Sky Lodge Hotel decided to transition to an IT-based solution. As a result, a web-based hotel reservation management system and accompanying website were designed and developed to deliver expected benefits to management, employees, and guests alike.

P. B. N. N. De Silva (2017). Web Site &Web Based Hotel Reservation Management System For Sky Lodge Hotel. University of Colombo School of Computing. BIT registration number: R100490 Index number: 1004905.

Development of a Web-Based Reservation Systems for Federal Polytechnic Kaltungo

According to Abdulrauf et al. (2023). Education is often regarded the most suitable tool for national development. This helps and assists citizens in developing the skills and attitudes required for nation building.

It is no longer news that many Nigerian university graduates and other higher educational institutions fall short of employer or industry standards [1]. Concerns have been raised over the years about Nigeria's hospitality/tourism education and training system as a source of skilled labor for the tourism industry.

To align with global best practices, the National Board for Technical Education (NBTE) mandated that all higher education tourism departments in the country must implement an operational reservation and booking mechanism for practical training. The department is tasked with equipping students with the essential skills and knowledge required to meet modern industry demands. However, persistent challenges in providing hands-on training for online reservation systems have resulted in a notable practical skills gap among graduating students entering the workforce.

To bridge this gap, this research report presents the development of an online reservation system that will be used to teach students of the Department of Tourism at Federal Polytechnic Kaltungo how to carry out online reservations of hotel rooms, airline

tickets, and food ordering. The system was designed to provide students with hands-on experience and practical skills in online reservation, which is an important aspect of the tourism industry.

Abdulrauf et al. (2023). Online reservation systems are web-based reservation management systems that allow customers to book and pay for online activities such as hotel rooms, tickets, and restaurant tables or related services through the website. This service is operated online, minimizing the personnel workload and the possibility of multiple reservations

Several Studies show that, online reservation systems were designed and developed to improve customer service, expedite the reservation process, and boost operational efficiency for merchants. Online reservation systems are used in a variety of businesses such as hospitality, transportation, and entertainment.

System development results demonstrated that the online reservation system created for the Department of Tourism at Federal Polytechnic Kaltungo effectively teaches students how to book hotel rooms, airline tickets, and food. Users found the platform to be highly user-friendly, featuring clear, easily understandable instructions alongside a visually appealing interface designed to enhance the user experience. To evaluate the system's quality, strength, and overall usability, the researchers conducted a preliminary study using a yes-or-no questionnaire. The evaluation process involved gathering direct feedback from the Dean of the School of Science, the Head of the Department, five instructors, and 13 students. This diverse group of participants was instructed to test every module within the system thoroughly before completing their feedback forms.

Abdulrauf et al. (2023). The development of the online reservation system at Federal Polytechnic Kaltungo marks a significant advancement in providing students with critical practical skills required by the modern tourism industry. Due to its user-friendly interface, clear instructions, and demonstrated instructional effectiveness, the platform serves as a highly valuable tool for teaching and learning within the department. Based on these successful outcomes, it is recommended that the department formally integrate this system into its curriculum for teaching reservation skills. To preserve its long-term effectiveness and usability, the software must also be regularly updated and properly maintained over time. Ultimately, further research is recommended to investigate how this system impacts graduate employability and the broader development of the tourism industry in Nigeria.

Abdulrauf, A., Ogar. A.U., Delmut, R,D., and Emmanuel, O, C. (2023). International Journal of Engineering and Information Systems (IJEAIS). ISSN: 2643-640X. Vol. 7 Issue 7. Pages: 25-38.

Understanding the Effects of Reservation Systems and Online Transaction Capacity on the Competitiveness of Travel Agencies

The study by Işkın, M., Prentice, C., Eker, N., and Şenge Ü. (2023). The purpose of this study is to determine the effect of travel agencies' reservation systems and online transaction capacities on their competitiveness. With the developments in technology and the integration of these developments in all business areas, travel agencies have also had to benefit from information and communication technologies. In the research, the primary data were used, and the data were collected by the questionnaire method.

The rapid acceleration of globalization and technology has driven profound socio-economic changes, giving rise to an "information society" where data is a highly valuable, tradeable product. To navigate this new landscape, modern global approaches emphasize the sustainable management of information and the strategic integration of technology.

Işkın, M., Prentice, C., Eker, N., and Şenge Ü. (2023). In the research, the primary data were used, and the data were collected by the questionnaire method. The convenience sampling method was used, and data were collected from a total of 566 travel agency managers in Istanbul. According to the results of the research, reservation systems and online transactions capacities of travel agencies affect their competitiveness. Reservation systems and online transactions capacities of travel agencies affect their competitiveness in specific issues such as cost, service quality and market share.

Although appropriate and enlightening results were obtained regarding travel agencies technological capacity and competitiveness, research was carried out on the perceptions and evaluations of the managers in this study. In this context, in future studies, studies can be conducted on the relationships between measurable quantitative variables such as the online transactions of travel agencies and the capacity and transaction volume of reservation systems and competition structures.

The study reveals that the traditional office travel agencies should be more successful than their competitors in terms of reservation systems and online.

Işkın, M., Prentice, C., Eker, N., and Şenge Ü. (2023). Understanding the Effects of Reservation Systems and Online Transaction Capacity on the Competitiveness of Travel Agencies. EJTHR 2024; 14(1): 55-70. https://doi.org/10.2478/ejthr-2024-0004.

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College of Information Technology Education

DEVELOPMENT AND IMPLEMENTATION OF A PYTHON-BASED HOTEL MANAGEMENT SYSTEM: A COMPREHENSIVE TOOL FOR ROOM RESERVATION, PAYMENT, AND ADMINISTRATION

The study by Alam, M., Rahman, M. S., Rivin,M.A. H., Uddin, M. B., & Sizan, M. M. H. (2024). The hotel industry faces challenges related to manual management processes, which can lead to inefficiencies and errors in operations such as booking, payment processing, and administrative tasks. To address these challenges, there is a growing demand for automated systems to streamline operations and improve overall efficiency.

This study investigates the design and implementation of a Python-based Hotel Management System built using the Django framework and MySQL database. The system is developed to handle essential hotel operations, including room booking, payment processing, and general management, while ensuring scalability, security, and usability.

The research follows a software development approach based on functional and non-functional requirements, with a focus on unit, integration, functional, and security testing to ensure the system's reliability and performance. Key features of the system include real-time booking functionality, secure payment processing, and authenticated user access for administrators and customers. The results reveal that the system effectively reduces operational errors, enhances the user experience for both hotel administrators and guests, and improves overall operational efficiency.

The goal of this study is to create an automated hotel management system using python to solve the operational inefficiencieswithin the hospitality industry involving rooms reservations, payment processing, and administrative management. The research addressed the scientific problem of how to combine advanced technology into a scalable, real-time, and user-friendlysolution for streamlining hotel operations.

The developed hotel management system successfully optimizes daily operations, enhances security, and improves decision-making through features like real-time updates and role-based access control. Thorough testing and a modular architecture ensure that the platform is reliable, scalable, easy to maintain, and adaptable to shifting technological trends. From a management perspective, the system provides the real-time data analysis necessary to boost customer satisfaction and streamline administrative workflows. Despite these strengths, the study is currently limited because the system was only evaluated under controlled conditions rather than in diverse, real-world environments. Furthermore, the current version focuses strictly on essential functionalities, leaving room for future enhancements like advanced analytics and third-party platform integrations. Future research aims to bridge these gaps by exploring dynamic pricing algorithms, IoT-enabled intelligent rooms, and large-scale implementations across different geographical regions.

References: Alam, M., Rahman, M. S., Rivin,M.A. H., Uddin, M. B., & Sizan, M. M. H. (2024). DEVELOPMENT AND IMPLEMENTATION OF A PYTHON-BASED HOTEL MANAGEMENT SYSTEM: A COMPREHENSIVE TOOL FOR ROOM RESERVATION, PAYMENT, AND ADMINISTRATION. American International Journal of Business and Management Studies 6(1), 15–44. https://doi.org/10.46545/aijbms.v6i1.322

2.2 Local Studies

Web-Based Reservation and Management System for Nature Spring Resort

According to Ramento, Dancel, Taberlo, and Anquillano. (2024). The significant improvements in accuracy, operational processes, and user contentment demonstrate the system's success.

The study focuses on the business which experienced challenges due to its manual reservation system that relied on logbooks, leading to errors and inefficiencies. To address these issues, the researchers developed a web-based reservation and management system that allows guests to book online and check availability.

As a result, they developed system transforms and improves the resort’s booking and management processes. This system provides a simple and user-friendly online interface for customers to book rooms and cottages effortlessly. Customers can also view the resort’s available amenities and services. The developed system includes real-time availability updates, assuring accurate booking information while reducing the possibility of overbooking. Furthermore, the management feature allows Nature Spring Resort owners to handle reservations and monitor guest preferences easily.

The development of the web-based reservation and management system for Nature Spring Resort has improved the reservation process, making it a valuable tool for both users and managers. By addressing and resolving past ineffectiveness, the system allows users to make bookings faster and more effective. The significant improvements in accuracy, operational processes, and user contentment demonstrate the system's success. These results illustrate the system's ability to improve the reservation and booking experience, making it more reliable and user-friendly. Overall, the developed web-based reservation and management system modernizes Nature Sprint Resort's booking process and significantly improves its operational capabilities.

Ramento, L. F., Dancel, D. M., Taberlo, E., & Anquillano, (2024) Web-Based Reservation and Management System for Nature Spring Resort. Journal of Pure and Applied Sciences. ISSN:2984-9896.

Online Reservation Inventory Systems in Higher Education: A Gap Analysis and Enhancement Framework with SMS Notification Support

The study of Sayson, J,M., Reyes, J,C., and Pisueña, J,S. (2025). Inventory and reservation management systems are essential for maintaining seamless administrative and academic operations within higher education institutions. As a core component of material management, inventory heavily impacts an organization's financial outcomes, operational performance, and customer satisfaction. To optimize these outcomes, universities have increasingly adopted digital inventory systems to improve the tracking, allocation, and utilization of campus resources. These tools are particularly valuable for managing shared assets like university equipment, facilities, and daily supplies. Ultimately, implementing these centralized digital platforms ensures the proper documentation necessary for strict institutional accountability and future planning.

Online inventory and reservation systems are widely implemented across universities, their efficiency and functionality vary significantly. Academic literature highlights that these digital systems are highly effective at eliminating manual errors, preventing untracked inventory, accelerating slow reservation processes, and providing crucial real-time updates. Most studies emphasize that successful systems share core features like item tracking, reservation logs, reporting functions, and user-level access controls. However, despite these technological advancements, current systems still suffer from notable limitations, including poor user notifications, system accessibility gaps, and limited integration with modern communication technologies.

Sayson, J,M., Reyes, J,C., and Pisueña, J,S. (2025). This research utilizes a systematic literature review and design science approach to analyze inventory and reservation systems within higher education institutions by evaluating published studies, reports, and case analyses. A primary focus of the study is examining the integration, or noticeable lack, of real-time SMS-based notification features. Implementing these text alerts can significantly improve user-administrator communication, reduce missed bookings, and enhance overall system responsiveness. By focusing on recurring limitations within Philippine state universities and similar campuses, the study identifies critical operational gaps and evaluates the distinct benefits of SMS integration. Ultimately, the insights gained from this literature will guide proposed system upgrades to support more efficient, user-friendly reservation and inventory processes.

This study successfully identified the key limitations of existing academic resource reservation systems and proposed a comprehensive enhancement framework to bridge these gaps. Developed through a rigorous literature review and design science research principles, the framework integrates vital features like real-time inventory tracking and optimized scheduling. It also introduces automated SMS notifications and secure, role-based access controls for users. By prioritizing usability and data-driven insights, the system aims to deliver a responsive and efficient experience for students, faculty, and administrators alike. Ultimately, this framework establishes a strong foundation for future system development that is highly scalable, configurable, and aligned with institutional needs.

Reference: Sayson, J,M., Reyes, J,C., and Pisueña, J,S. (2025). Online Reservation Inventory Systems in Higher Education: A Gap Analysis and Enhancement Framework with SMS Notification Support. International Journal of Research in Engineering and Science (IJRES). ISSN (Online): 2320-936 4, ISSN (Print): 2320-9356. www.ijres.org Volume 13 Issue 7 ǁ July 2025 ǁ PP. 73-82

Digital Transformation of Reservation Services: Development of the Foxrock Internet Reservation Management System

The study by Cajetas et al. (2026). The rapid advancement of technology has fundamentally transformed the way organizations manage reservations and deliver customer services. A growing number of businesses have adopted digital reservation platforms to streamline the booking process and reduce turnaround time. Such systems offer customers a significantly more convenient and accessible channel for scheduling appointments or securing reservations without the constraints of location or business hours, which in turn contributes to higher levels of customer satisfaction. Moreover, by automating booking transactions, these platforms substantially reduce the volume of incoming phone calls and email inquiries, allowing staff to redirect their attention toward higher-value operational tasks

A Management Information System (MIS) serves as a unified platform that gathers, processes, stores, and distributes data to facilitate decision-making, organizational coordination, and operational oversight. The present study examines how MIS contributes to improving the efficiency of organizational decision-making by delivering timely data and in-depth analytical insights that align with institutional objectives while harnessing emerging technologies such as artificial intelligence and cloud infrastructure. By bringing together human resources, data assets, hardware, software systems, and established procedures, MIS enables organizations to streamline operations and empowers managers to arrive at well-informed strategic and day-today decisions.

The online reservation platform developed by the research team is expected to assist Foxrock Internet Services in modernizing and enhancing its existing service delivery mechanisms. Through the company's userfriendly website, clients will be able to conveniently schedule internet installation appointments entirely online. Foxrock Internet Services, founded in 2018 by Mr. Kevin Paul Pacina and Mrs. Sharmaine Duraca Pacina, operates in Barangay Malbago, Madridejos, Cebu, and offers both fiber and wireless internet connectivity to its subscribers.

Numerous rural and underserved localities continue to suffer from poor or completely absent internet connectivity, commonly referred to as "dead zones." This lack of access severely curtails residents' capacity to engage with online services and leaves them uninformed about available connection options, underscoring the persistent digital divide separating rural communities from urban counterparts.

Cajetas et al. (2026). Prior research indicates that digital systems strengthen communication channels between service providers and clients while alleviating administrative burdens. By offering real-time availability information and automated booking confirmations, these platforms eliminate the need for repetitive follow-up inquiries. This automation significantly cuts the time and financial costs traditionally associated with manual scheduling workflows. In a standard system setup, customers simply supply their personal details, product preferences, and preferred payment option. Finally, settlement can be conveniently handled by field personnel or facilitated through flexible digital services like GCash to increase accessibility.

Cajetas et al. (2026). Organizations increasingly adopt digital reservation platforms to optimize their workflows and improve operational efficiency. These modern systems reduce dependence on physical paperwork while minimizing scheduling conflicts like duplicate bookings. Additionally, they offer users the flexibility to submit reservations at any hour without being restricted by standard office hours. Despite these clear advantages, a number of organizations still rely on traditional manual booking methods that remain prone to errors. In contrast, the Foxrock Internet Reservation Management System addresses these inefficiencies through clearly identified functional requirements and an intuitive, user-centered design.

The overall findings of the study reflect a highly favorable reception of the Foxrock Internet Reservation Management System across all evaluated dimensions. Respondents assigned exceptional ratings ranging from "Very Satisfactory" to "Excellent" across a broad array of core functional criteria. These positive results confirm that the system successfully met, and in several respects surpassed, user expectations regarding its primary capabilities. The only identified area with notable potential for future growth was the responsiveness of the mobile application interface. Although the mobile experience still achieved a "Very Satisfactory" rating of 4.3, this specific score indicates a clear opportunity to further optimize performance and user engagement.

Reference: Cajetas, M, M, Jr., Alo, S, M., Chavez, C., Ilustrisimo, D., Arranchado, K, M., Cabahug, J, Jr., and Alegre, K, B. (2026). Digital Transformation of Reservation Services: Development of the Foxrock Internet Reservation Management System. International Journal of Computer Science and Mobile Computing, Vol.15. pg. 51-59. DOI: https://doi.org/10.47760/ijcsmc.2026.v15i03.006.

Development and evaluation of a web-based retail management system for small business operations

Dela Cruz and Santos (2022) investigated the development of a web-based retail management system designed specifically for small businesses. In their study, they observed how shifting from manual transaction and inventory processes to an automated web-based system directly improved operational performance. Their findings showed that

the system significantly enhanced transaction accuracy by minimizing errors commonly associated with manual data entry and recordkeeping. Additionally, the system enabled real-time inventory tracking, helping business owners monitor stock levels instantly and make informed replenishment decisions. Such automation reduced discrepancies between

recorded and actual inventory, thereby enhancing overall business productivity and reliability.

These results are relevant to the current study because they demonstrate how automation and integration of digital systems can improve efficiency, accuracy, and data visibility — principles that support the need for an automated study hub management system with features like real-time scheduling, resource tracking, and user interactions.

Dela Cruz, J. R., & Santos, A. (2022). Development and evaluation of a web-based retail management system for small business operations. Journal of Small Business Information Systems, 6(4), 23–35.

SORMS: Steffi's Online Reservation Management System

The Study of Balintag et al. (2025). SORMS (Steffi's Online Reservation Management System) is a highly successful, web-based digital solution developed

specifically to transition Steffi's Garden Resort away from the inefficiencies of a manual booking system. concerns about usability, functionality, safety, privacy, and accuracy of information, manual reservations remain prone to errors, often causing inconvenience for customers.

These inefficiencies highlight the necessity for service providers, such as hotels and resorts, to implement online reservation management systems. Research suggests that hoteliers or staff can opt to pay only the software's license cost, which is more cost-effective than paying commissions to external booking portals when customers make reservations. Additionally, customers find it more convenient to book at their leisure and prefer utilizing technology over manual data entry.

Manual reservation procedures in various industries have proven inefficient. In restaurants, customers often experience long wait times for table availability, making software-based systems a more effective alternative.

Developing an online reservation system requires the establishment of digital infrastructure, including servers, networks, and software applications, contributing to the goals of Sustainable Development Goal (SDG)9—resilient infrastructure, sustainable industrialization and innovation. Through digitalization, businesses can optimize operations and free up resources for further innovation.

Businesses must adapt to technological advancements not only to stay competitive but also to enhance service quality. This study aims to develop an efficient and user-friendly reservation system for Steffi’s Garden Resort, ensuring a seamless experience for both customers and resort staff. While reservations may seem like a straightforward task, they involve numerous challenges that can impact guest satisfaction.

To address these challenges, the proponents developed the Online Reservation Management System—a user-friendly platform that allows customers to seamlessly book and manage their reservations online, reducing errors and enhancing overall guest satisfaction.

The study developed Steffi's Online Reservation Management System (SORMS) using Agile Scrum methodology, with researchers implementing key phases including product backlog refinement, sprint planning, and iterative development cycles. The researcher employed standard design tools - Gantt charts, data flow diagrams, and entity-relationship models - to create a robust architecture, while maintaining close collaboration with resort staff and customers to ensure practical relevance.

References: Balintag, E, E., Concepcion, K, J, H., Dagman, P, J., Ducos,. J, A, Fernandez, D,. and Villanueva E. (2025). SORMS: STEFFI'S ONLINE RESERVATION MANAGEMENT SYSTEM. PSYCHOLOGY AND EDUCATION: A MULTIDISCIPLINARY JOURNAL. Volume: 36. Pages: 783-790. ISSN 2822-4353.

Managing Customer Reservations of BulSU Hostel through the Development of Online Information and Reservation System

This study by Castro et al. (2014), later published as a book in 2017, offers practical insights into how technology can streamline service operations and improve customer experiences in institutional facilities. Released initially in the International Journal of Advances in Management and Economics by Kiban Research Publication Pvt. Ltd. and subsequently by LAP LAMBERT Academic Publishing, the authors explain that an online information and reservation system acts as a centralized platform that connects customers, staff, and facility resources to automate booking processes, ensure real-time availability tracking, and simplify administrative tasks.

This concept is highly relevant to the development of various facility management systems—such as library room reservation or hostel booking platforms—as such systems require effective integration of user access, resource monitoring, and service coordination to reduce errors, save time, and enhance overall operational efficiency. Accurate and timely reservations are necessary to ensure proper service and positive customer experience for many service-oriented businesses.

In this study, the proponents considered factors like customer satisfaction and positive experience upon the stay in a hostel which started in the reservation procedure. In order to manage customer reservations of Bulacan State University (BulSU) Hostel, the proponents design and develop an “Online Information and Reservation System for BulSU Hostel”. Different functionalities were incorporated into the developed system like the creation of different user account for the staff/front desk and online customers, prices updates through online admin portal, payments thru bank account, confirmation thru email after reservation, report generation concerning income, number of customers and monthly report, counting the numbers of viewers and settings for rates, taxes and other discounts. Visual Basic .Net and ASP. Net. were used for the system front-end while Microsoft Access was used as the database application as well Crystal Report was used to display reports.

The developed system was evaluated using the software quality standard ISO 9026 with different software criteria as follows: Functionality, Accuracy, Reliability, User-friendliness, and Security and all of them were interpreted as “Acceptable” based on the equivalent ratings presented in the Likert scale. Reserving a room ensures or guarantees the guest about the availability of a room on arrival at the hotel, as reservation is a commitment made by the hotel, when the hotel has accepted the reservation request Prior to internet services, travelers could write, telephone the hotel directly, or use a travel agent to make a reservation. Online hotel reservations are also helpful for making last minute travel arrangements. Hotels may drop the price of a room if some rooms are still available. In Bulacan State University (BulSU), Philippines, Hostel started in 2005 which caters the growing number of clientele who wants a comfortable and convenient place to stay while taking short-term courses in the University or attending conferences or seminars within the campus. The BulSU hostel is a budget-oriented, shared-room accommodation that accepts individual travelers or groups for short-term stays. In some countries the word hostel can also refer to student accommodation that provides common areas and communal facilities.

The University Hostel is also popular among foreign students and guests of the University. The Hostel 45 rooms as well the facilities are also made available for the interests of the students of the university. Students taking up Hotel and Restaurant Administration likewise use the facility for their on-the-job training. Usability and User Experience The study “Managing Customer Reservations of BulSU Hostel through the Development of Online Information and Reservation System” by Castro et al. (2014) highlights that usability and user experience are central to the success of web-based reservation platforms. The authors emphasize that the system was designed with a simple, clear interface and straightforward navigation, allowing both customers and staff to perform tasks easily even with limited technical knowledge.

Evaluation results showed that the system was rated highly in terms of ease of use, understandability, and responsiveness, ensuring that users can quickly find information, complete bookings, and access services without confusion. Supported by related local studies such as Vega and Generoso II (2025) and Dacanay et al. (2023), which also stress the importance of intuitive design and user-centered features, these findings confirm that prioritizing usability and positive user experience not only increases user acceptance and satisfaction but also ensures that the system serves its purpose effectively by being accessible, practical, and enjoyable to use for all types of user.

References: Castro, M. D. B., Nuqui, A. V., Custodio, E. B., & Bernardino, I. S. (2014). Managing Customer Reservations of BulSU Hostel through the Development of Online Information and Reservation System. International Journal of Advances in Management and Economics, 3(5). Kiban Research Publication Pvt. Ltd.; ISSN: 2278-3369 Castro, M. D. B. (2017). Managing Customer Reservations of BulSU Hostel. LAP LAMBERT Academic Publishing; ISBN: 9783330343948

WEB-BASED HOTEL RESERVATION SYSTEM

The Study by Olaño, J., Agcaoili, A., Canlas, D., Narbasa, I,C., and Ordillas, J,M. (2025). Due to enormous increase in tourism worldwide, standards especially to small establishments, have improved considerably. Tourists in various markets depend on well-organized reservation system with ease in accessibility which aids in ensuring well planned trip whether business or pleasure. The aim of this study is to help Queens Hotel in assessment of their reservation and billing

Web-Based Hotel Reservation System can be used to serve as more reliable and speed up the process of Recording and Estimating the total cost of customers reserve room. They save you the burden of learning and understanding complex reservation and billing process. This study is to show the advantages and disadvantages that have in the Web-Based Hotel Reservation System. Reservation software completes Estimating the total cost of customers reserve room it would take to do them manually, whilst your reservation staff might not like it.

The purpose of hotel reservation system is to provide a place for the travelers to sleep. A hotel reservation system is also known as central reservation system (CRS). The central reservation system is an assistance for hoteliers to manage all of their marketing sales to see the availabilities to be seen all sales channels such as travel agencies

References: Olaño, J., Agcaoili, A., Canlas, D., Narbasa, I,C., and Ordillas, J,M. (2025). The Faculty of Bataan Heroes Memorial College.

Booking and Reservation System: A Unified Application Using Location-Based Services for Sustainable Tourism Networks

According to Gosela, R,R., and Encarnacion, R,E. (2024). The LakByahe application is a unified booking and reservation system that addresses tourism challenges in the Municipality of Nasipit in Agusan del Norte. That aims to provide a dedicated web-based platform as the current travel industry rely on social media for updates because there is a lack of organized source of information for local insights.

The LakByahe Application offers a ground-breaking solution to revolutionize tourism in Nasipit, Agusan del Norte, Philippines. It is a unified application aiming to address existing challenges by incorporating user-centric features. Through an advanced location-based services and streamlined booking processes, the application intends to enhance efficiency, accessibility, and sustainability in the tourism sector. By providing real-time tracking, efficient transit solutions, and a comprehensive directory of establishments.

The application can fill information gaps by acting as a dedicated source. It enhances travel experiences with user-friendly features, real-time tracking, and efficient transit solutions, aligning with modern trends. Additionally, it fosters economic growth by connecting tourists with local businesses. The app aims to revolutionize travel and transportation booking, offering a seamless rental experience and overcoming tourist difficulties in these municipalities

The researchers successfully identified gaps in localized tourism solutions and crafted a platform specifically tailored to the distinct needs of tourists in Nasipit. The proposed LakByahe App represents a pioneering effort to transform the local tourism landscape through innovative technology. By addressing current system shortcomings and incorporating user-centric features, the application aims to improve efficiency, enhance accessibility, and foster sustainable tourism networks. It effectively tackles challenges related to user experience and reliability by integrating advanced location-based services, streamlining the booking process, and ensuring robust data security. Furthermore, the app's focus on sustainability promotes equitable tourism practices that directly support under-resourced areas. Stakeholders are also provided with an interactive dashboard, a comprehensive establishment directory, and monthly statistical reports to continuously monitor and enhance the tourism network. Ultimately, this comprehensive approach optimizes the booking and reservation system while setting a robust framework for future digital advancements in Nasipit.

Reference: Gosela, R,R., and Encarnacion, R,E. (2024). Booking and Reservation System: A Unified Application Using Location-Based Services for Sustainable Tourism Networks. International Journal of Advanced Research in Science, Communication and Technology (IJARSCT). ISSN (Online) 2581-9429. Volume 4. DOI: 10.48175/IJARSCT-18889.

Effects of Computer Reservation System in the Operations of Travel Agencies

The Study by Felicen, S, S., and Ylagan, A, P. (2016). In travel industry, the main tool used the computerized booking systems known as Global Distribution Systems or GDS. In reservations for travel product, the use of GDS effectively is the key skill of Travel Industry employees. In Australia and New Zealand, Amadeus and Sabre reservation systems are used daily by travel agency, airline, and tour company employees. Being competent in these skills will help graduates land his first job in travel.

GDS or CRS system is for purely (information transfer) logistical functions. They store current information about all available service providers and have the necessary infrastructure to transfer such data. This means that the system also performs additional tasks related to service distribution, which in the area of goods is typically carried out by freight forwarders. They support the transport of goods by eliminating the physical distance between the producer and the sales mediator or the consumer respectively. CRS's are a combination of infrastructure measures offered to interested providers in the tourist industry.

The continuous growth of 24/7 internet connectivity has transformed the tourism industry by making online buying and selling ubiquitous. This shift led to the disintermediation of the traditional travel services model, forcing legacy service providers and Global Distribution Systems (GDS) to migrate to new digital models.

While consumers increasingly prioritize a convenient, comfortable, and burden-free purchasing experience, experts note that the immense scope of technology still comes with distinct limitations. Consequently, the field of travel services research in India remains a vast, open area ripe for mutually beneficial collaborations between industry and academia.

The researchers chose this topic to determine the effect of using Computer Reservation System among Travel Agencies in terms of technical, human and financial aspect. This will help the Internship office to include the identified travel agencies in their linkages where the students will be deployed for internship. The result of this study will also be helpful and can be utilized in the course travel and tour operations with computer reservation system.

Felicen, S, S., and Ylagan, A, P. (2016). This study aimed to assess Computer Reservation System used in the travel agencies in Batangas City and Lipa City, Philippines. Specifically, it determined the profile of the travel agencies in Batangas, identify the reservation system used by travel agencies, find out the services offered by travel agencies; determine the features of the CRS; and determine the effects of using computer reservation system in terms of human aspect, technical aspect and financial aspect.

Felicen, S, S., and Ylagan, A, P. (2016). A majority of the surveyed travel agencies in Batangas and Lipa City have been operating for 6 to 10 years, employ 3 to 4 people, and utilize the Abacus computer reservation system (CRS) for ticketing, reservations, and tour packaging. This CRS enhances agency efficiency and productivity by securing airline data, managing inventories, and connecting guests to all forms of travel. While the study concludes that technology brings both positive and negative impacts to the workplace, these findings will be integrated into curriculum development and pre-orientation programs to better prepare internship students. However, due to the study's limited geographic scope, future research should expand to include other variables and a wider respondent pool across the CALABARZON region.

Felicen, S, S., and Ylagan, A, P. (2016). Effects of Computer Reservation System in the Operations of Travel Agencies. Asia Pacific Journal of Multidisciplinary Research Vol. 4 No.4, 23-28. E-ISSN 2350-8442.

2.3 Related Literature

DESIGNING A WEB-BASED RESTAURANT RESERVATION INFORMATION SYSTEM WITH REQUIREMENT PROTOTYPING METHOD

According to Sutjiadi, R., Rahmawati, T., Kristianto, A., and Kanessa, F, T. (2026). The development of information technology is proliferating, along with the increasing human need for fast, precise, and accurate information. The role of information technology in supporting the business world will make it easier for business people to run their business.

Internet-based digital communication tools are currently increasing. This cannot be separated from several supporting factors, including the increasingly affordable price of digital communication devices, such as smartphones, tablets and laptops. In addition, internet users are currently also experiencing a relatively rapid increase.

The use of information technology in the business world also brings more or less changes in the company's business strategy. Computerised methods make recording data and information more systematised and accessible for further processing. This disruption will change the company's marketing strategy from conventional to digital through information technology and Internet media (Khairunnisa, 2022). With the application of digital marketing, business actors can develop a broader market and add value to increase customer satisfaction, which also impacts increasing company turnover (Saputra et al., 2023).

One of the business fields that can utilise information technology is restaurants. In the past, restaurant business actors still used manual records to run their business, such as recording customer orders, reservations, and financial transactions. This manual recording certainly has risks such as data loss, recording errors due to human error, and slow processing of the data. Currently, by utilising information technology infrastructure, data is recorded centrally through a database, where the data can be recalled quickly, system validation to minimise recording errors, and processed more rapidly into reports, which is helpful for further business analysis. Business analysis with computerised data can increase the company's productivity (Ahmad et al., 2022).

The Wisata Kampung Kemiri information system integrates table reservations and food ordering into a single, comprehensive web platform. This digital design ensures the system is easily accessible while providing real-time data regarding available tables and menu orders. Rigorous black-box testing and User Acceptance Testing (UAT) validated the performance of these reservation and ordering features. Additionally, a well-designed user interface helps consumers navigate the platform quickly, which increases user engagement and overall convenience. By employing the Requirement Prototyping software development method, the system was able to significantly accommodate specific user needs. This flexible approach allowed users to participate actively in the development process, ensuring the final product fully met their expectations. In the future, this application could be developed into a mobile app that is easier to use on customers' smartphones. Additionally, adding features such as reward points and online payments through a payment gateway would provide added value for customers.

References: Sutjiadi, R., Rahmawati, T., Kristianto, A., and Kanessa, F, T. (2026). DESIGNING A WEB-BASED RESTAURANT RESERVATION INFORMATION SYSTEM WITH REQUIREMENT PROTOTYPING METHOD. Techno Nusa Mandiri: Journal of Computing and Information Technology. P-ISSN: 1978-2136 | E-ISSN: 2527-676X. Vol. 22. DOI https://doi.org/10.33480/techno.v22i1.6110.

Operational Efficiency in Retail: Using Data Analytics to Optimize Inventory and Supply Chain Management

The study of Famoti et al. (2025). Operational Efficiency in Retail: Using Data Analytics to Optimize Inventory and Supply Chain Management. The retail industry faces constant challenges in balancing customer demand with cost-minimization during inventory and supply chain management. To enhance operational efficiency, data-driven approaches leverage historical sales data, seasonal trends, and machine learning to improve demand forecasting accuracy. Retailers can maintain optimal stock levels and mitigate overstocking or stockout risks by applying techniques like ABC analysis, safety stock calculations, and Economic Order Quantity (EOQ) optimization. Furthermore, data analytics streamlines supply chain logistics through vendor scorecards that track supplier performance, alongside geographic data used for real-time shipment monitoring and route optimization.

Famoti et al. (2025). Integrating advanced warehouse management systems and robust inventory software further streamlines retail operations. These modern analytics platforms equip retailers with data visualization, reporting tools, and predictive analytics to empower informed, data-driven decision-making. Real-world case studies across both large-scale and small-to-medium enterprises demonstrate the tangible benefits and best practices of successfully adopting these data analytics tools.

Key components of operational efficiency in retail include Inventory Management. Efficiently managing inventory is essential for meeting customer demand while minimizing carrying costs. Retailers need to strike a balance between stocking enough inventory to fulfill orders promptly and avoiding excess inventory that ties up capital and storage space.

Data analytics has become indispensable for modern retail operations by driving operational efficiency, cost savings, and enhanced customer experiences. By analyzing historical data, seasonal trends, and external factors, retailers can achieve accurate demand forecasts that align inventory levels with market demand while preventing stockouts or overstocking. Furthermore, these data-driven strategies optimize supplier relationship management through performance tracking while utilizing techniques like ABC analysis, safety stock calculations, and Economic Order Quantity (EOQ) to minimize carrying costs. Famoti et al. (2025).

Logistics optimization through route planning, real-time tracking, and warehouse management system integration leads to faster, more reliable, and cost-effective supply chain operations. Additionally, leveraging data on customer preferences, purchase history, and behavior allows retailers to deliver personalized experiences that enhance loyalty, satisfaction, and sales. Ultimately, data analytics replaces intuition with solid evidence, providing actionable insights that support successful strategic decision-making across pricing, product assortments, and marketing campaigns.

Famoti et al. (2025). In conclusion, the future of retail operations relies heavily on the effective utilization of data analytics. Retailers who embrace data-driven decision-making will enhance their operational efficiency and position themselves for long-term success in a rapidly evolving market. By investing in the right infrastructure, tools, and institutional culture, businesses can unlock the full potential of these analytics to achieve significant competitive advantages. Therefore, the time to act is now for retailers to adopt these data-driven strategies and transform their operations for a prosperous future.

References: Famoti, O., Ezechi, O, N., Ewim, C, P, M., Eloho, O., Ajayi, T, P, M., Igwe, A, N., and Ihechere, A, O. (2025). Operational Efficiency in Retail: Using Data Analytics to Optimize Inventory and Supply Chain Management. International Journal of Scientific Research in Computer Science, Engineering and Information Technology. ISSN : 2456-3307. Volume 11, Issue 1. Page 1483-1494. doi : https://doi.org/10.32628/CSEIT251112173

System Development Life Cycle (SDLC)

According to Roger S. Pressman and Maxim (2019) in their book Software Engineering: A Practitioner's Approach, structured methodologies like the System Development Life Cycle (SDLC) provide a step-by-step framework for system development. Applying to SDLC promotes quality, reduces risks, and ensures that all user requirements are adequately addressed.

The book stated above is relevant to the proponents' study as it provides the necessary framework for the systematic design and implementation of the study hub's features. By following the SDLC, the proponents can ensure that the real-time scheduling, billing, and reporting modules are thoroughly tested and functional before deployment.

References: Pressman, R. S., & Maxim, B. R. (2019). Software engineering: A practitioner's approach (9th ed.). McGraw-Hill Education.

IOT Environmental-monitoring System Development

According to Hur,B (2021) Mosquitoes pose a major threat to humans and animals by transmitting dangerous viruses and parasites like West Nile, dengue, Zika, and malaria. Consequently, research and population control efforts are vital to saving lives from these vector-borne diseases. Researchers frequently analyze climate and environmental data to better understand mosquito breeding habits and manage their populations. However, broad macro weather data often fails to provide accurate information for specific, localized areas. Therefore, monitoring precise microclimate conditions is necessary to achieve the high-resolution data and deep analysis needed for effective control.

For the measurement and monitoring of microclimate environmental data, the development of a custom environmental monitoring system for mosquito research has been in progress by the multidisciplinary research team from three departments: Entomology, Engineering Technology, and Industrial and Systems Engineering at Texas A&M University. An Engineering Technology team at Texas A&M University has been working on the engineering aspect of the research project.

In order to expand and broaden the impact of this research as well as on engineering education, three capstone teams were formed to develop parts of the IOT (Internet of Things) environmental monitoring system for Mosquito research. A total of 12 undergrad engineering students have been working on these capstone projects. Each capstone team is composed of four undergraduate students. In addition, two graduate students have been involved as graduate student mentors to assist this effort.

The names of the three capstone teams are Big Dog Engineering (Team 1), AIM-N (Team 2), and Pond Hoppers (Team 3). Big Dog Engineering is developing a mobile weather station that can autonomously navigate to specific locations to collect climate data and return to base. AIM-N is building a customized IOT network and a low-cost cluster server using Raspberry Pi modules to process sensor data and perform machine learning analysis. Pond Hoppers is constructing a specialized quadcopter drone capable of floating on standing water to collect samples and monitor breeding site conditions. Together, these three interconnected projects aim to provide high-resolution, mobile environmental data specifically geared toward advanced mosquito research and population control.

Hur,B (2021) To combat fatal vector-borne diseases, researchers integrated their research and teaching efforts to develop a customized IOT monitoring system for mosquito control. This initiative led to the creation of three distinct capstone projects launched in Fall 2020. Student teams were tasked with designing a mobile weather station, a Raspberry Pi cluster server, and a water-sampling drone. Although the three teams focused on significantly different engineering tasks, they discovered strong, beneficial synergies by working under this common theme. By the conclusion of the projects in Spring 2021, all three teams successfully completed their individual goals. Ultimately, their success proved that integrating these distinct systems was possible, and the authors now plan to continue advancing this technology for future mosquito research.

References: Hur,B (2021). IOT Environmental-monitoring System Development for Mosquito Research Through Capstone Project Integration in Engineering Technology. American Society for Engineering Education. Paper ID #34086.

Reservation System and Customer Management Optimization

According to Siregar, Gading, & Raya, (2026) The entertainment industry, especially karaoke services, is one of the sectors that continues to grow along with the increasing need for recreational activities. At many karaoke venues, the room reservation process is still done manually using book notes or simple forms. This practice often causes various operational problems, such as inaccurate data recording, schedule clashes due to lack of visibility into room availability, cost calculation errors, and difficulties in tracking transaction history. These problems not only hinder the effectiveness of operational staff but also have an impact on the quality of service felt by customers. In addition, manual recording has the risk of data loss, writing errors, and a lack of integration between the information needed in management decision-making.

Seeing these conditions, the use of computer-based information systems is one of the relevant solutions to overcome reservation management problems in karaoke businesses. Desktop-based reservation systems allow for structured data storage, easy access to information, and increased efficiency in administrative processes. This digital approach can provide real-time visibility into the status of the room, minimize potential schedule clashes, and speed up the service process. Thus, the implementation of a computerized reservation system can provide added value for business actors, both in terms of operations and customer satisfaction.

This research demonstrates that the application of information technology in karaoke reservation management can significantly enhance operational performance. By replacing manual processes with a computerized system, businesses can achieve greater efficiency, transparency, and reliability in their services. The findings of this study not only provide practical benefits for karaoke operators but also contribute academically by offering a structured model for developing reservation systems in similar service industries. Future research may expand on this work by incorporating online reservation features, payment gateways, or data analytics modules to further improve system functionality and business competitiveness.

When compared to the manual reservation process, the developed system provides significant improvements, especially in terms of speed, accuracy, and data consistency. The time required to complete a reservation transaction is shorter, as staff no longer need to make manual records or check room availability one by one. In addition, data digitization has proven to minimize recording errors, such as customer name errors, data duplication, or schedule inconsistencies. Full integration between the customer, room, and transaction modules makes it easier for officers to monitor all operational activities.

References: Ridho Fadlan Fahira Siregar, Rafli Arya Gading, Fathul Hady Raya (2026). Karaoke Reservation System for Room and Customer Management Optimization. Journal of Artificial Intelligence and Digital Business (RIGGS). P-ISSN: 2963-9298, e-ISSN: 2963-914X.

Web Development Technologies

According to documentation from MySQL and W3Schools (2023), modern web applications rely on a combination of database management systems (MySQL) and server side scripting (PHP). MySQL is used for storing and retrieving structured data efficiently, while PHP enables dynamic content generation and secure transaction handling.

This information is essential to the proponents as it justifies the choice of the technology stack to be used. Using these tools allows developers to build a system that is scalable, secure, and accessible across devices.

References: W3Schools. (2023). PHP, HTML, CSS, JavaScript tutorials. https://www.w3schools.com MySQL. (2023). MySQL documentation. https://dev.mysql.com/doc/

Development Student-Teacher Online Booking Appointment System in Academic Institutions

According to Bello, R. O., Olugbebi, M., Babatunde, A. O., Bello, B. O., and Bello, S. I. (2016), the study titled "Student-Teacher Online Booking Appointment System in Academic Institutions" was published in the Journal of Computer Science and Control Systems (Volume 9, Issue 2, ISSN 2315-9430). The research explains that traditional manual appointment methods are often time-consuming, disorganized, and prone to conflicts, so they developed a web-based system using PHP, JavaScript, and MySQL to address these issues. The platform allows students to view teachers' available time slots, book, reschedule, or cancel appointments anytime, while teachers can manage their schedules and approve requests easily. It is designed to be user-friendly, accessible on different devices, and capable of providing real-time updates—ultimately making the scheduling process faster, more convenient, and more efficient for both students and educators in academic settings.

Bello et al. (2016) also highlighted that the primary goal of developing such systems is to make appointment processes less time-consuming, easier to navigate, and more user-friendly through simple, click-based operations. The system follows a two-tier client-server architecture that communicates via the Hypertext Transfer Protocol (HTTP), allowing the web server to handle core functions such as user authentication, page generation, and database management. In terms of functionality, students can register, log in, book appointments, and send messages stating the purpose of their visit, while teachers can create schedules, approve or decline booking requests, view messages, and manage all appointments efficiently. By centralizing these processes and ensuring data is securely stored and easily retrievable through the database, the system ultimately aims to minimize waiting times and maximize the number of students that can be served effectively — contributing to better service delivery and operational efficiency in educational institutions.

References: Bello, R. O., Olugbebi, M., Babatunde, A. O., Bello, B. O., & Bello, S. I. (2016). Student-Teacher Online Booking Appointment System in Academic Institutions. Journal of Computer Science and Control Systems, 9(2). Published in 2016; Computer Science and Control Systems; ISSN: 2315-9430

Vega, R., & Generoso II, J. (2025). Ereserve: An Online Facility Reservation System of One Philippine Higher Education Institution. Philippine Journal of Information and Communication Technology. Published in 2025; Philippine Society of Information Technologists; ISSN: 2945-387X

Development of Online Student Course Registration System

According to Singh et al. (2016), manual student course registration processes are labor-intensive, time-consuming, and require students to complete multiple forms and obtain physical signatures from teachers, advisors, deans, and accounting staff. To address these challenges, the authors developed a web-based system using PHP, jQuery, Apache, and MySQL that automates the entire workflow. The system is divided into five main modules — Admin, Masters, Transactions, Reports, and Utilities — and supports different user roles including students, faculty, heads of department, registrar, accounts officer, and dean. It allows students to register for courses from any location, enables officials to review and approve requests digitally, maintains data consistency and security through role-based access, and eliminates duplication and errors common in manual record-keeping. The study concludes that the online system is more secure, user-friendly, and efficient, significantly reducing workload for both students and institutional staff while making registration faster and more accessible.

With the advent of Information Technology in the last decade, the major focus has shifted from manual systems to computerized systems. Various systems viz. railway reservation, hospital management etc. involving manual work have been automated efficiently. Student course registration process in colleges involve filling registration forms manually, getting it signed by respective subject teachers, and then getting the documents acknowledged from the concerned Advisors, College Deans and Accounts Officers respectively. Finally the registration forms are submitted in the Administrative Branch. As is evident, this process is very laborious and time consuming. An Online Student Course Registration System has been developed to simplify the current manual procedure. This system has been developed using PHP, jQuery, Apache and MySQL. The front-end is designed using PHP with excerpts of code written using jQuery and back-end is designed and managed through MySQL. This system software is more secured, user-friendly and less time-consuming.

Singh et al. (2016) further noted that web-based applications provide distinct advantages over traditional methods, including the elimination of physical presence requirements, reduction of paperwork, minimization of data duplication, and real-time accessibility of information. By centralizing records in a structured database, institutions can maintain data consistency, integrity, and security, while generating accurate reports to support decision-making and resource planning. The study highlights that implementing such systems not only reduces the workload of both students and administrative personnel but also enhances transparency, accountability, and the overall quality of academic services. It concludes that online course registration systems are essential tools for modern educational institutions, enabling them to manage academic processes effectively, save valuable time and resources, and adapt to the growing demand for digital solutions in education.

References: Singh, R., Singh, R., Kaur, H., & Gupta, O. P. (2016). Development of Online Student Course Registration System. Oriental Journal of Computer Science and Technology, 9(2). Oriental Scientific Publishing Co., India; ISSN: 0974-6471; DOI: 10.13005/ojcst/9.02.02

AN ANALYSIS OF RESTAURANT TABLE RESERVATION SYSTEMS

The study by Mahajan, K., Pati, M., and Patel, J. (2025). Presenting the design and implementation of a restaurant table reservation system with the increasing demand for online services. Efficient table reservation system have become essential for both customer and restaurant owners our system aims to streamline the reservation process, enhance customer experience, and optimize restaurant operations.

The system incorporates various features, including a user-friendly interface for customer to browse available tables, select preferred dining times and make reservation the system utilizes database management to store and retrieve reservation information securely to develop the system, we employed python programming language along with relevant libraries such as web development.

The dining industry is increasingly adopting technology-driven solutions like automated table reservation systems to optimize operational efficiency and enhance the customer experience. These systems stand out as pivotal tools that allow restaurants to streamline booking processes, manage seating arrangements, and improve service quality. This research paper explores the significance of digital reservation systems within the contemporary restaurant landscape through a comprehensive review of existing literature, industry reports, and case studies. Ultimately, the study aims to shed light on the benefits, challenges, and implications of adopting these systems for various stakeholders, including restaurant owners, staff, and customers.

Mahajan, K., Pati, M., and Patel, J. (2025). The reservation system influences the overall dining experiences for customer, including case of booking, customization options, and communication with the restaurant considering factors such as increased table utilization it may include features such as managing multiple reservation, handling cancellations. This project may involve developing a userfriendly interface for customers to view availability select a desired time and date, provide contact information, and receive confirmation.

The restaurant table reservation project offers significant, long-term benefits to both customers and owners. Through the implementation of this efficient system, customers enjoy greater convenience, reduced waiting times, and an enhanced overall dining experience. Concurrently, restaurant owners are able to optimize table turnover, improve resource allocation, and ultimately increase their profitability. Moving forward, continuous monitoring and regular updates to the system will be necessary to ensure it remains highly effective over time. This ongoing maintenance will allow the platform to keep pace with evolving customer needs and changing industry demands.

References: Mahajan, K., Pati, M., and Patel, J. (2025). AN ANALYSIS OF RESTAURANT TABLE RESERVATION SYSTEMS. INTERNATIONAL JOURNAL OF PROGRESSIVE RESEARCH IN ENGINEERING MANAGEMENT AND SCIENCE (IJPREMS). e-ISSN : 2583-1062. pp: 1018-1021.

IMPACT OF BUSINESS INTELLIGENCE AND ANALYTICS ON DECISIONMAKING IN ONLINE RESERVATION SYSTEMS WITHIN THE HOSPITALITY SECTOR

The Study by Ukandu, O., Falana, T., Adiu, A., Kanu, R., and Ebiesuwa, S. (2025). This study investigates the impact of business intelligence and analytics (BIA) on decision-

making in online reservation systems within the hospitality sector. This study presents a comprehensive review of ten research papers in the field of BIA within the hospitality sector. The findings indicate that organizations effectively leveraging BIA experience enhanced decision-making, improved operational efficiency, revenue growth, and customer satisfaction. Integration of BIA leads to improved customer experiences through personalization and recommendation systems. Adopting BIA tools results in significant operational efficiency gains by addressing issues in real-time.

The hospitality sector is a dynamic and expanding industry that encompasses a wide range of businesses, including hotels, restaurants, resorts, and tourism services. This sector greatly impacts the global economy and is driven by a rising demand for travel, leisure activities, and technological advancements. To remain competitive in a crowded marketplace, businesses within this industry must prioritize providing exceptional customer experiences and delivering high-quality services. Successful organizations understand that achieving operational efficiency and profitability relies heavily on adapting to changing trends and embracing innovation. Ultimately, these factors have led the hospitality sector to experience notable and transformative changes over the years.

Technological advancements have revolutionized various aspects of the industry, from online reservation systems and mobile applications to personalized marketing and guest engagement. Digital platforms and social media have empowered consumers to share their experiences, shaping the reputation and success of hospitality businesses. As the hospitality sector continues to evolve, organizations are increasingly turning to data-driven decision-making to gain a competitive advantage. Business intelligence and analytics (BIA) have emerged as powerful tools that enable hospitality businesses to extract valuable insights from vast amounts of data

Ukandu, O., Falana, T., Adiu, A., Kanu, R., and Ebiesuwa, S. (2025). This study highlights the powerful impact of business intelligence and analytics on decisionmaking in online reservation systems. The results demonstrate that organizations that effectively leverage datadriven insights are able to make informed decisions, enhance customer experiences, improve operational efficiency, and drive revenue growth. By adopting appropriate business intelligence tools, integrating data from multiple sources, and employing advanced analytics techniques, organizations can unlock the full potential of their online reservation systems.

Ukandu, O., Falana, T., Adiu, A., Kanu, R., and Ebiesuwa, S. (2025). IMPACT OF BUSINESS INTELLIGENCE AND ANALYTICS ON DECISIONMAKING IN ONLINE RESERVATION SYSTEMS WITHIN THE HOSPITALITY SECTOR. Indian Journal of Computer Science and Engineering (IJCSE). e-ISSN : 0976-5166. pp.640 - 651.

Reservation systems as a tool of tourist services marketing

The study by Myskiv, G. and Wojtan, S,N. (2022). Modern reservation systems was conducted to determine their role in the tourism industry and to improve understanding of the basics of marketing activities of tourism enterprises. The article aims to analyze the functioning of modern reservation systems in the market of tourist services, to structure the existing reservation systems according to the operation levels and according to the distribution channels of the tourist product.

To remain relevant in a fiercely competitive market, any tourism company must continuously improve its business activities. Meeting the constantly growing requirements of modern customers makes it essential for these companies to have permanent access to up-to-date market information. To achieve this, businesses need to invest in reliable, multifunctional reservation systems that can effectively track customer needs and preferences. Implementing these systems ultimately helps companies attract additional clients and successfully expand their existing distribution channels. Furthermore, this technological investment enables active, high-quality promotion of tourist services across the broader market.

Reservation systems play an extremely important marketing role in the tourism industry by providing travelers with free access to select, book, and cancel hotels, restaurants, or flights. These integrated digital platforms also significantly increase tourist attraction by offering simplified, secure payment methods. Because reservation technologies are developing rapidly alongside shifting market demands, there is a critical need to continuously study modern global booking networks. This research is highly relevant as it explores how these systems interact, coordinate with one another, and ultimately impact overall company competitiveness. Ultimately, analyzing these modern platforms helps tourism businesses better promote their products and successfully deliver high-quality hospitality services.

The global and domestic tourism industries are actively developing, with computer reservation systems serving as the foundational basis for this growth. Driven by the widespread digitalization of the service sector, these systems simplify how tourists access, select, and book various hospitality services. Ultimately, these multifunctional platforms allow tourism companies to operate with high efficiency, cover vast geographic regions, and successfully attract millions of travelers.

Myskiv, G. and Wojtan, S,N. (2022). This study analyzed the modern global travel reservation market to successfully map out its distinct hierarchical structure. The foundation of this system hierarchy is determined by the geographic scale of coverage and the specific distribution channels utilized for tourist services. At the top tiers, Global Distribution Systems (GDS) occupy the first global level, while Online Travel Agencies (OTA) and Restaurant Reservation Systems (RRS) comprise the second regional level. The remaining framework descends from the third level's corporate Central Reservation Systems (CRS) down to local automated websites at the fourth level, and finally to local non-automated call centers at the fifth level.

The extreme ease of booking through various computerized reservation systems successfully attracts a growing number of tourists to both established and newly opened hospitality venues. These digital platforms also allow travelers to effortlessly secure essential transit services, including airline flights, railway transport, and car rentals. Ultimately, reservation systems have evolved into a vital tool for distributing and promoting tourism, which comprehensively drives the marketing development of modern travel companies.

Reference: Myskiv, G., & Nycz-Wojtan, S. (2022). Reservation systems as a tool of tourist services marketing. Economics, Entrepreneurship, Management, 9(2), 7-18.

Hotel Reservation Management System

The study by Shun, K,W., and Hassim Y,M,M. (2021). Malaysia is a growing tourist destination, there has been a good rise in the number of hotels and resorts in Malaysia due to the broadening of the tourist sector. The rapid development and commercialization of Information and Communication Technologies (ICT) for travel and tourism industry have prompted hotels and other enterprises in this sector to increasingly adopt technologiesinto their businesses.

Hotel Time Johor Bahru is one of the hotel in Malaysia where located in Johor Bahru, Johor. The hotel currently uses BlogSpot blog for displaying hotel's details only without having a proper reservation system and relied on third-party hotel reservation platform. The problems faced by the hotel currently are there will be fewer profits earned since the hotel needs to pay the commission to the external booking portal eac tim a guest made a reservation from the respective booking portal. Besides, the hotel looks less impressive without having its own booking engine since some guests prefer the hotel's own booking system for reserving a room instead of using an external booking portal.

The objectives of this project is to design a hotel reservation management system and then to develop a hotel reservation management system for Hotel Time Johor Bahru. In this project, the Waterfall model is applied to assist the project development. The system is a web-based system developed based on Structured approach.

The Hotel Reservation Management System implemented for Hotel Time Johor Bahru automates major operations such as booking, accommodations, accounts, inventory, and reporting, while offering online reservation tracking for rooms and halls. To support these operations, the platform integrates five distinct domain modules management, reporting, inventory, reservation, reward, and registration tailored for use across five specific user roles, including administrators, multiple staff tiers, and guests.

The Hotel Reservation Management System allows traveling or local guests to easily compare rates, review services, and securely book rooms online without visiting the property beforehand. Simultaneously, hotel staff can effortlessly manage room inventories, update pricing, process transactions, and access daily financial reports for data analysis. Finally, the platform enables owners to promote their business directly by creating custom marketing packages and promotional deals to attract more customers.

By receiving direct online reservations, hotel owners can bypass costly Online Travel Agency (OTA) commission fees, allowing those savings to go directly to the bottom line as increased profit while enabling closer guest engagement. Furthermore, this user-friendly system reduces the risk of lost bookings, making the reservation process easier for customers and helping hotel staff perform their duties more effectively.

Shun, K,W., and Hassim Y,M,M. (2021). The Hotel Time Johor Bahru Reservation Management System is a web-based platform that boosts operational efficiency for hoteliers while providing a convenient room booking process for guests. By utilizing this software, owners avoid paying continuous commissions to third-party portals and only need to cover the basic license cost. However, a major disadvantage of the current system is the lack of a channel manager to connect the platform with external booking networks. This limitation prevents real-time data synchronization during reservations, which can ultimately lead to accidental overbooking issues. To resolve this, future developments should integrate a channel manager for synchronous updates, alongside mobile application support and features like meal or group reservation management.

Reference: Shun, K,W., and Hassim Y,M,M. (2021). Hotel Reservation Management System. Faculty of Computer Science and Information Technology, Universiti Tun Hussein Onn Malaysia, 86400, Johor, MALAYSIA. Vol. 2 No. 2 (2021) p. 973-992. DOI: https://doi.org/10.30880/aitcs.2021.02.02.061.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local studies indicate that web-based management systems significantly enhance operational efficiency, improve data accuracy, and increase customer satisfaction. The literature also stresses the importance of structured software development methodologies, real-time data processing, and user-centered interface design.

However, most existing management systems lack the specific integration needed for a study hub environment, such as the simultaneous tracking of membership duration, room occupancy, and food inventory.

The present study addresses these gaps by designing and developing an Integrated Study Hub Management System for WS Students & Professionals Lounge. This system integrates reservation management, automated membership tracking, and inventory control into one platform. This integrated approach ensures improved efficiency, accuracy, and usability for both the staff and the customers.

2.5 Conceptual Framework

INPUT	PROCESS	OUTPUT
User Information Details: User data such as name, contact number, address, and account credentials. Reservation and Membership Data: Reservation requests, selected room rates/services, membership details, and booking schedules. System Records: Existing records of users, reservations, memberships, billing transactions, sales reports, and room availability.	User Authentication and Access Control: This process verifies user login credentials and applies Role-Based Access Control (RBAC) to ensure that only authorized users can access specific system functions and data. Reservation Management: Validating schedule availability to prevent double-bookings Membership and Time Usage Monitoring Process: This process manages membership status, tracks check-in and check-out times, calculates usage duration or expiration, and supports accurate billing computation. Billing and Sales Report Generation Process: The system processes billing transactions, records payments, and automatically generates summarized sales reports for monitoring daily operations and transactions.	Real-Time Status Updates: Displays updated reservation schedules, room availability, membership status, and billing information. Organized Digital Records: Securely stored records of users, reservations, memberships, billing transactions, and sales data for easy retrieval and monitoring. Sales Reports: Summarized reports of transactions, reservations, and billing activities to support administrative monitoring and decision-making.

Figure 1. Conceptual Framework of an Integrated Study Hub Management System for WS Students & Professionals Lounge

The conceptual framework of an Integrated Study Hub Management System for WS Students & Professionals Lounge aims to automate manual workflows to improve efficiency, accuracy, and data security by ensuring that every transaction is properly secured and recorded. The inputs include customer or user information, service requests, and existing records such as reservation details, memberships, and user data. These inputs are processed through core system functions such as user authentication, reservation management, membership handling, time-based usage session tracking, billing, and sales report generation.

The process begins when a Student/Professional or an Admin/Staff logs into the system. The system validates credentials against the Users table and applies Role-Based Access Control (RBAC) to determine access permissions.

The reservation phase wherein the user or staff checks the availability of rooms or tables. After the verification, the reservation request is submitted, and the system will verify the availability to prevent double bookings. Once validated, the system updates the reservations table and sends a booking confirmation.

The Check in, upon physical arrival, the staff triggers a check-in, recording the time_in in the time logs. When the user leaves, the time_out is recorded, and the system calculates the total stay duration. At the end of the day, the system harvests all transaction data. Then, the system summarizes high-level metrics, including total_check_ins, total_logins, and total_time_logged into the Daily Reports table. The Admin reviews these reports to monitor daily performance and usage patterns.

Through these processes, the system ensures accurate scheduling, prevents double bookings, and efficiently manages transactions. The outputs generated include real-time status updates, well-organized records, and summarized sales reports on daily operations. Overall, the framework demonstrates how the proposed system converts raw data into useful information to enhance efficiency, accuracy, and decision-making within the study hub.

Chapter 3

3.1 Methodology

In this project, the proponents will focus on how the management, staff, and customers of WS Students & Professionals Lounge – La Paz Branch handle their daily operations to better understand their workflow. By combining different data gathering methods, the proponents will be able to collect useful and meaningful insights. These findings will help the proponents solve relevant issues such as manual reservation errors, difficulties in tracking memberships, and inaccuracies in billing and sales recording.

The proponents will conduct surveys and interviews to give them a better understanding of the manual tools the lounge currently uses to manage their data. This will also help the proponents see the bigger picture, such as how often issues like double bookings or stock discrepancies occur and how the staff feel about the current paper-based system. Based on these results, the proponents can gather data that will assist in designing and developing an Integrated Study Hub Management System that specifically meets the needs of the WS Students & Professionals Lounge.

The surveys will be used to determine the frequency of operational problems and the difficulties faced by the staff during peak hours. This will help the proponents understand the user experience and the specific pain points of the current system. The collected data will provide a basis for the proponents to assess and provide solutions to the challenges and concerns faced by the lounge administrator and employees.

Furthermore, the proponents will also perform interviews and direct observations to better know the daily routine of the lounge's staff. Knowing the challenges through face-to-face interactions will help the proponents identify necessary improvements in the reservations, memberships, billing, and sales report process. The combination of these research methods will allow the proponents to gather both quantitative and qualitative results, combining technical information with the personal experiences of the users to help design and create the proposed management system.

3.2 Methodology Table

METHODS	PARTICIPANTS	PURPOSE
Surveys	WS Students & Professionals Lounge Owner, General Manager, Staff, and Regular Customers	To gather quantitative data regarding the frequency of manual errors and the level of satisfaction with the current system.
Interviews	WS Students & Professionals Lounge Administrator and Employees	To obtain qualitative insights into the specific operational challenges, pain points, and desired features for the new system.
		To observe the actual workflow of reservation, membership management,

Observations	WS Students & Professionals Lounge Staff and Customers	time room usage, and billing to identify bottlenecks in the manual process.
Data Analysis	The Proponents	To synthesize the collected data, identifying patterns in operational errors, and correlating them with system requirements.
Solution Development	The Proponents	To design and develop the Integrated Study Hub Management System based on the analyzed needs and technical specifications.

Table 1. Methodology Table

3.3 Research and Development Stage

The Research and Development (R&D) stage is essential in guiding the Integrated Study Hub Management System from conceptualization to full implementation. This stage is dedicated to ensuring that the system specifically addresses the unique challenges of WS Students & Professionals Lounge. The process begins with extensive research to understand the current manual workflow and the frustrations of the staff regarding data integrity and customer service speed.

The proponents will analyze existing management systems to identify industry best practices while focusing on the specific needs of a study lounge environment. Through data-gathering methods such as surveys, interviews, and observations, the proponents will document the risks associated with the current manual setup, including issues in user registration, reservation handling, membership management, time-based room usage tracking (duration or expiration), billing, and sales report generation. This data will serve as the foundation for defining the functional requirements of the system.

After the data-gathering phase, the proponents will transition to the Design Stage. In this phase, the proponents will create the system's architecture, focusing on key features such as user registration, reservation bookings, membership management, time-based room usage tracking (duration or expiration), billing, and sales report generation. The goal is to develop a user-friendly interface that simplifies the staff's workload. Throughout this process, the proponents will use the initial feedback to ensure the design remains intuitive and responsive to the users' needs.

Following the design, the proponents will proceed to the Development phase. This is where the actual coding and building of the system take place. The proponents will focus on creating a functional prototype that integrates the front-end interface with a secure back-end database. The development will be iterative, meaning the proponents will continuously refine the features based on preliminary internal testing to ensure the system is stable and efficient.

Once the prototype is fully developed, it will enter the Testing Phase. The system will undergo rigorous testing, including functionality testing, data integrity checks, and User Acceptance Testing (UAT). The proponents will involve the WS Students & Professionals Lounge staff to test the system in a controlled environment to ensure it can handle real-world scenarios like peak-hour bookings. Any bugs or vulnerabilities identified during this phase will be rectified before the final deployment.

Lastly, the R&D process concludes with Evaluation. The proponents will review the overall performance of the system against the initial objectives. By analyzing the final test results and user feedback, the proponents will confirm if the system effectively solves the identified problems of manual errors and tracking difficulties. Once all requirements are satisfied, the Integrated Study Hub Management System will be ready for full implementation, providing a secure and organized digital environment for the lounge.

3.4 Research Design

This study uses a developmental research design because the main goal of the proponents is to design and build a functional system that will solve the operational problems of WS Students & Professionals Lounge. The proponents will focus on creating a tool that is practical and can be used in the real-world setting of the business.

A descriptive approach is also used to help the proponents understand the current situation of the lounge. This involves looking into how the staff currently work, identifying the common problems they encounter, and knowing what specific features are needed for the new system. By describing the existing manual process, the proponents can better design a solution that fits the needs of the lounge administrator, staff, and customers.

3.5 System Development Methodology

The project follows the process of the System Development Life Cycle (SDLC) methodology, following these phases:

- **Planning** – In this phase, the proponents will identify the problems in the manual operations of WS Students & Professionals Lounge, such as reservation errors and the difficulty of tracking memberships, billing, and daily sales report. The proponents will also define the system requirements and scope to make sure the project addresses the issues in manual recording, inaccurate data handling, and the slow process of managing customer information.

- **Analysis** – The proponents will gather data through surveys, interviews, and direct observations with the administrators, staff, and customers of WS Students & Professionals Lounge. This includes analyzing the risks of using paper-based logs, such as data loss, double bookings, and the time-consuming process of manually checking and updating records, including reservation details and billing transactions.

- **Design** – The proponents will create a simple and user-friendly interface that is easy for the lounge staff to navigate. This phase also includes designing a database that can securely store customer information, membership records, reservation details, room rates/services, billing data, and sales reports to make data retrieval much faster.

- **Implementation** – The proponents will begin developing the system and setting up the necessary database. Key features will be implemented, including user registration, real-time reservation management, an automated membership tracker, time-based room usage monitoring (duration or expiration), billing transactions, and sales report generation. Security features such as user login accounts and activity logs will also be included to make sure that the lounge data is protected and only authorized staff can access it.

- **Testing** – The proponents will check all the features of the system to make sure there are no bugs or errors. They will test if the registration, reservation, membership, billing, and sales report modules are working correctly. The staff of WS Students & Professionals Lounge will also be asked to try the system to see if it meets their expectations and if it is easy enough for them to use during their daily operations.

- **Deployment** – In this phase, the proponents will launch the system for actual use in the WS Students & Professionals Lounge. The proponents will also provide training for the employees to teach them how to use the system confidently and safely. The system will be monitored to ensure it runs smoothly, and the proponents will be ready to answer any questions from the staff to guarantee a good user experience.

3.6 Agile Development Methodology

Figure 2. Agile Methodology

The Agile Methodology is chosen for the Integrated Study Hub Management System because the project aims to develop a secured, organized, and user-friendly system for the WS Students & Professionals Lounge. Agile is flexible and adaptable to changing requirements, which is important since the needs for managing user registration, reservations, membership tracking, time-based room usage, billing transactions, and sales reporting may be refined during development. This methodology allows the proponents to receive continuous feedback and opinions from the lounge management, making sure that the final system effectively addresses their manual operational issues. It also promotes strong collaboration between the proponents and the client, ensuring that their expectations are met and all technical challenges are identified and resolved early.

In the Requirements Analysis phase, the proponents will gather detailed information about the current manual processes of the lounge. This includes identifying specific problems such as double bookings, manual inventory errors, and the difficulty of tracking membership duration. The proponents will define the system requirements and scope to ensure that the proposed system directly addresses the operational challenges of the lounge staff and administrator.

During the System Design phase, the proponents will plan and design the overall architecture of the system, including its features and layout. This involves creating a user-friendly interface for the reservation, membership management, and billing modules, as well as designing a robust database to securely store user information, membership details, reservation data, time-based room usage, billing transactions, and sales reports securely. The design will focus on making the system intuitive for the staff to minimize the learning curve and ensure a smooth transition from paper-based processes to a digital system.

In the Development phase, which involves coding and implementation, the proponents will build the system based on the finalized design. The core features, such as user registration, real-time reservation management, automated membership tracking, time-based room usage monitoring (duration or expiration), billing transactions, and sales report generation, will be programmed and integrated. The proponents will ensure that the technology used is stable and capable of handling multiple transactions simultaneously without having data errors or system delays.

For the Testing phase, the proponents will conduct rigorous evaluation to ensure that the system functions properly and is free from critical errors. This includes verifying the accuracy of billing calculations, the reliability of reservation management, and the effectiveness of the preventing scheduling conflicts feature. The staff and administrator of WS Students & Professionals Lounge will also be invited to test the system prototype to determine whether it meets their requirements and to identify any necessary improvements before final implementation.

Lastly, in the Deployment phase, the proponents will present the system as a fully functional prototype for the WS Students & Professionals Lounge. This includes conducting a basic orientation or training session for the staff to ensure they know how to use the system effectively and confidently. The proponents will also monitor the initial usage of the system to guarantee a smooth user experience and gather final feedback to ensure that the system is fully ready for the lounge’s daily operations.

3.7 Data Collection Techniques

To design and develop an Integrated Study Hub Management System that truly meets the needs of the management, staff, and customers of WS Students & Professionals Lounge, the proponents will gather important data and insights using different techniques and methods. These techniques will help the proponents ensure that the system is user-friendly, efficient, and aligned with the problems and issues that need to be solved.

1. Surveys

The proponents will distribute surveys to the target users, including the lounge owner, general manager, staff, and employees. These surveys will help the proponents understand how they currently handle reservations, membership management, billing transactions, and sales reports, as well as the common problems they encounter and their expectations for a more organized digital system. The survey questions will also cover usability, system efficiency, and preferred features, which will guide the proponents in further development of the system.

2. Interviews

The proponents will conduct one-on-one interviews with the lounge administrator and selected staff members who are involved in managing daily operations, including reservations, billing, and record-keeping. This will allow the proponents to gain a deeper understanding of the challenges faced by the lounge in more detail—information that a simple survey might not provide. The proponents will ask the interviewees to share their personal experiences, concerns, and ideas on how to improve the efficiency and reliability of the reservation, billing, and management processes.

3. Focus Groups

The proponents will also organize a small group discussion with the lounge staff and manager to further discuss the proposed management system. This discussion will allow each participant to share their opinions, suggestions, and concerns regarding the planned features. By doing this, the proponents can identify which concerns are the most important and prioritize which features should be developed first to better meet the user's needs.

4. Observations

To better understand the daily workflow, the proponents will conduct direct observations of the existing manual processes being used in the lounge. This includes monitoring how the users' or members' information is recorded, how reservations are made, how the availability of study pods is checked, how the time duration/expiration of every customer is used, how the billing is transacted, and how the sales report is tracked. This will help the proponents spot the exact parts of the process where delays and errors usually happen.

5. Document Analysis

The proponents will also review existing records, logs, and articles related to study hub management and POS (Point of Sale) systems. This includes examining past reservation logs, billing records, sales report lists, and relevant studies on automated management systems. Through this analysis, the proponents will gain a clearer understanding of current practices and identify best practices that can be applied in developing a system tailored specifically to the needs of the WS Students & Professionals Lounge.

3.8 System Architecture and Design

Overview

The Integrated Study Hub Management System is designed to make the daily operations of WS Students & Professionals Lounge more organized, efficient, and digital-friendly. The system is built to solve the problems of manual recording, double bookings, and the hard way of tracking memberships, billing transactions, and sales reports. By moving from a paper-based system to a digital one, the lounge can ensure that all data is kept in a safe and organized place, making it easier for the staff to manage the hub and for the customers to enjoy the services.

The system supports different types of users, specifically the Admin, Staff, and Customers, each having their own roles and duties. The Admin has full control, such as managing user accounts and reservations, viewing detailed sales reports, and tracking the overall operations of the lounge. The Staff can handle the real-time reservations, process users or members information details, check room availability status, and billing transactions. Meanwhile, the Customers can easily view available study pods and manage their reservations or bookings. This organized flow ensures that the lounge's operation is smooth and that errors are avoided.

3.9 System Architecture

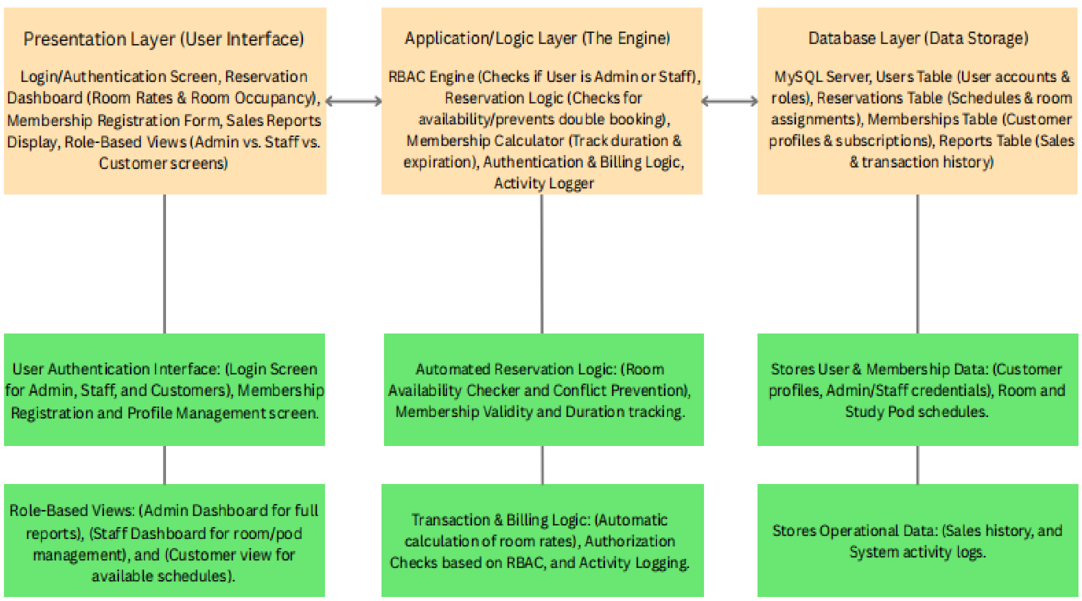
The Integrated Study Hub Management System will be developed using a Three-Tier Architecture to ensure a structured, scalable, and organized flow of data. This architecture consists of the following three layers:

- **Presentation Layer** – This layer provides the user interface (UI) where administrators, staff, and customers can interact with the system. It features a responsive dashboard for user registration, monitoring room occupancy, making reservations, and processing membership. For the administrator and staff, it includes management panels for handling users, reservations, memberships, transaction records, and real-time sales reporting. This layer also uses Role-Based Access Control (RBAC) to ensure that the interface and menus are customized based on the user's role—restricting sensitive management functions to authorized personnel while providing a simple, user-friendly interface for customers to register, make reservations, and view relevant account or billing information.

- **Application or Logic Layer** – This serves as the "brain" of the system that handles all user requests and processes the core business logic. It includes the User Management module for handling user registration and authentication, the Reservation Module that checks for room availability and prevents double bookings, the Membership Module that manages membership status, calculates usage duration, and subscription validity, the Billing Module that processes and calculates payments accurately, and the Sales Report Module that generates transaction summaries. This layer ensures that all operations are validated and correctly processed before sending or retrieving data from the database and returning results to the presentation layer.

- **Database Layer** – This layer is responsible for the secure and organized storage of all essential information using MySQL. It stores user profiles, membership records, transaction logs, room schedules, billing transactions, and sales reports. The database layer ensures data integrity, consistency, and security, making sure that every transaction and system activity is recorded accurately. It also allows fast and reliable data retrieval for real-time processing, reporting, and auditing purposes.

3.9.1 System Architecture Diagram



System Architecture Diagram

Figure 3. System Architecture Diagram of Integrated Study Hub Management System

3.10 System Components and Modules

The Integrated Study Hub Management System is made up of different modules that work together to create a simple and organized management tool. It uses automated tracking and role-based access to make sure that the operations are smooth and secured from errors or unauthorized access.

User Management Module

This manages the login and profiles of the admin, staff, and customers. Its key features include secure login for different user levels, profile management for members, and password management to keep accounts safe. It also handles session management to make sure that only authorized staff can access the management dashboard.

Reservation Booking Module

This is the core module for managing study pods and rooms. Its key features include a real-time availability checker to avoid double bookings, a walk-in/reservation form for customers and staff, and a schedule monitor that shows which rooms are occupied or vacant. It also tracks the time duration of stay to help the staff manage the turnaround of customers.

Membership Subscription Module

This module handles the records of regular members of the lounge. Its key features include a registration system for new members, tracking of membership expiration dates, and a history of their visits. It helps the management identify loyal customers and manage special discounts or membership perks.

Sales Reports Module

This provides the tools to manage the sales, transaction history, and other supplies in the lounge. Its key features include an automated sales and tracker that records every transaction. This ensures that the inventory is always accurate and easy to audit.

Admin Dashboard and Reporting Module

This provides the administrator with a centralized tool to see the overall status of the business. Its key features include viewing sales reports (daily, weekly, or monthly), monitoring user activities, and managing the list of room rates. It also provides analytic reports on peak hours to help the manager decide on staffing needs.

Notification and Alert Module

This module helps in communicating important updates to the staff and customers. Its key features include alerts for successful bookings, notifications when a customer's time is almost up, and system alerts if there are issues with the data records.

Security and Access Control Module

This part includes all the key security features of the Integrated Study Hub Management System:

- Use password encryption to protect the login details of the admin, staff, and customers.
- Implements Role-Based Access Control (RBAC) so that staff can only access the user's or members' details, reservation bookings, and billing while the admin can access the sensitive sales reports.
- Ensures that all reservation and membership data are backed up and safe from being lost or deleted.
- Protects the customer information from unauthorized access or data breaches.

3.11 System Architecture and Design

3.11.1 Context Diagram

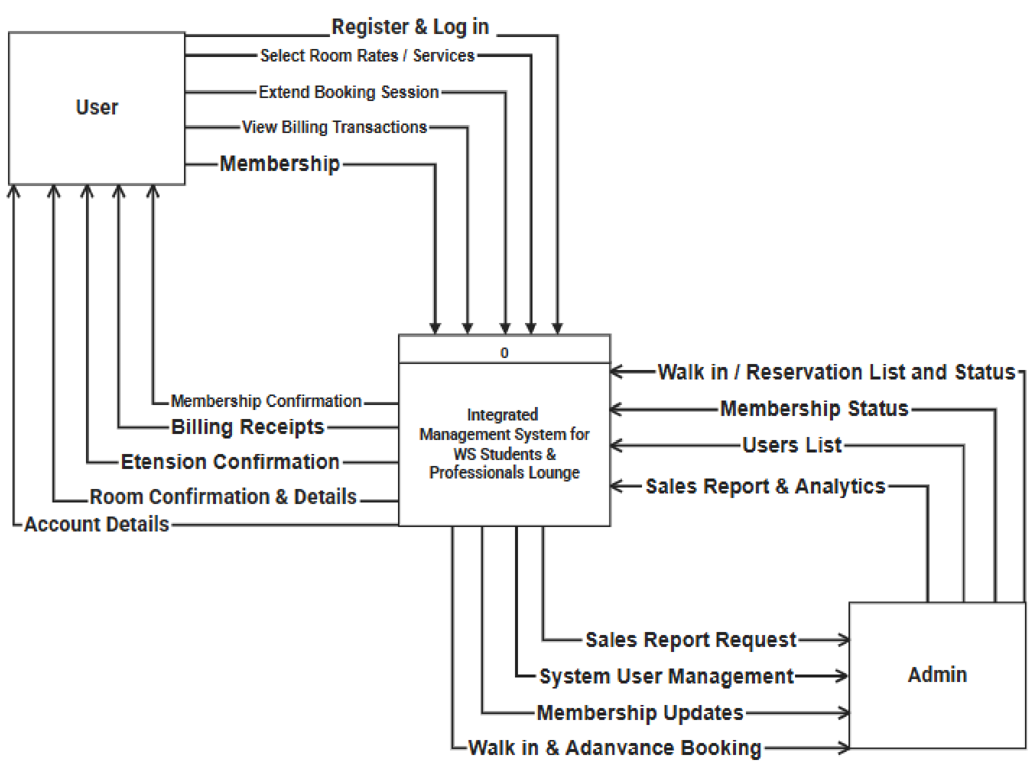


Figure 4. Context Diagram of an Integrated Study Hub Management System for WS Students & Professionals Lounge

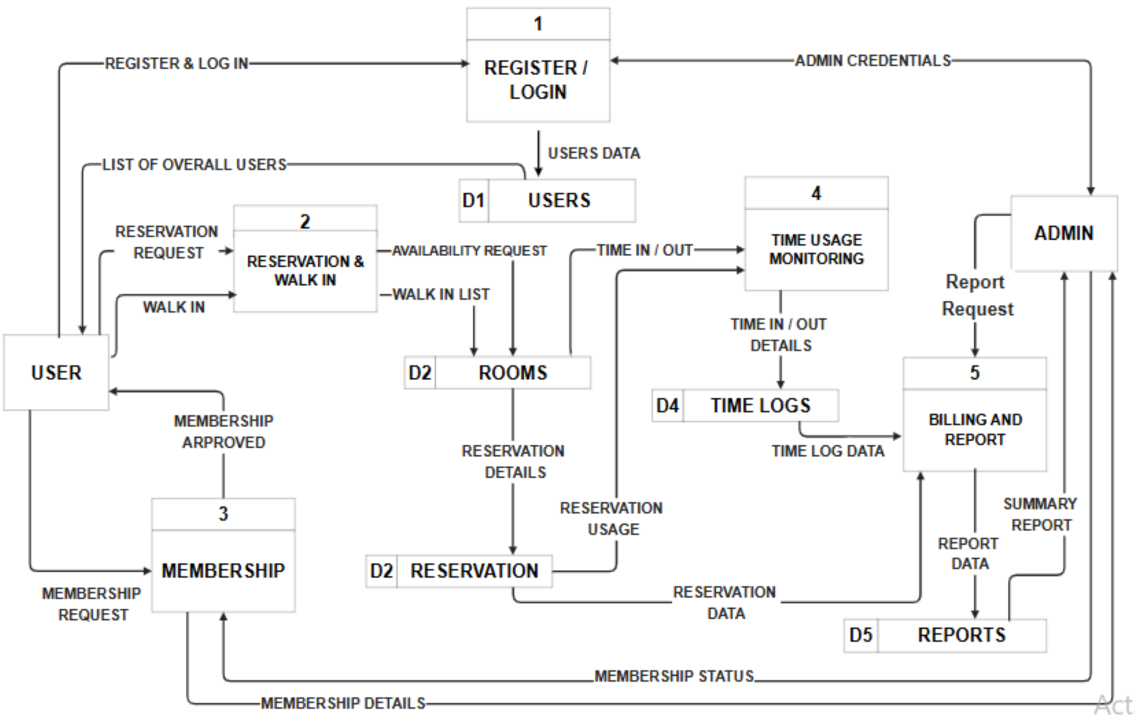


Figure 5. Data Flow Diagram of an Integrated Study Hub Management System for WS Students & Professionals Lounge

Significance of Context and Data Flow Diagram

The hierarchical structure of the context diagram and the Data Flow Diagrams (DFDs) provides a detailed blueprint of the Integrated Study Hub Management System for WS Students & Professionals Lounge. It outlines the flow of information from the conceptual overview down to the specific operational processes, ensuring that every transaction—from room bookings to sales report updates—is properly tracked and organized.

As shown in Figure 4. Context Diagram, the system’s scope is established by outlining the primary interactions between the external entities and the core system. The User initiates the flow by submitting registration requests, login credentials, room booking reservations, and requests to extend booking sessions. In return, the system provides confirmations, room details, and billing receipts. Simultaneously, the Admin interacts with the system by managing room rates, monitoring customer walk-ins, and performing system user management. The Admin also receives critical outputs such as sales reports and analytic through this centralized flow.

Moving to the Level 1 DFD, these high-level interactions are decomposed into five discrete sub-functions: Register/Login, Reservation & Walk-in, Membership, Time Usage Monitoring, and Billing and Report. For example, when a user makes a reservation, the system checks for availability against the Rooms (D2) data store. The Time Usage Monitoring process tracks "Time In/Out" details, which are then recorded in the Time Logs (D4) to ensure accurate duration tracking.

The Level 2 and Level 3 DFDs provide the most detailed view of the system's internal actions, such as the automatic calculation of total billing based on the data retrieved from time logs and reservation usage. This detailed modeling approach helps in design validation and ensures that all technical and operational requirements of the WS Students & Professionals Lounge are met, providing a smooth and error-free experience for everyone.

3.11 Use-Case Diagram

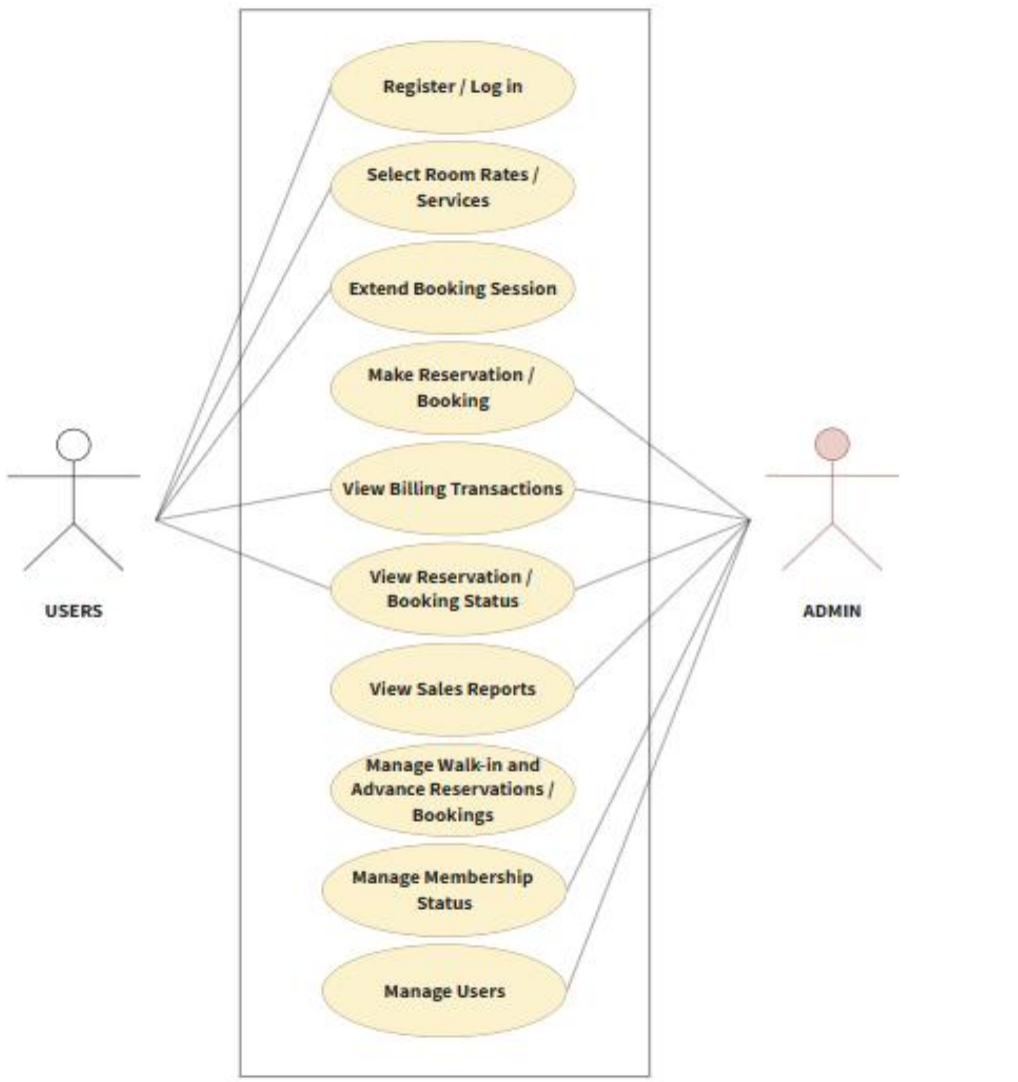


Figure 6. Use-Case Diagram of an Integrated Study Hub Management System for WS Students &u Professionals Lounge

Significance of Use-Case Diagram

The Use-Case Diagram for the Integrated Study Hub Management System is essential as it illustrates the dynamic behavior of the platform by defining the interactions between the users and the various functions of the system. It establishes a structured and secure approach to lounge management by mapping out the specific tasks that the two primary actors—the User (Students/Professionals) and the Admin (Staff/Management)—can perform within the system's boundary. It presents the main functions available within the system and shows how each actor interacts with these features to support the daily operations of WS Students & Professionals Lounge. By defining these interactions, the diagram helps ensure that all system requirements are clearly identified and properly organized according to user roles and responsibilities.

From the users' perspective, the system provides a simple and user-friendly process for accessing study hub services. Users can Register/Log In, Select Room Rates/Services, Make Reservations/Bookings, Extend Booking Sessions, View Reservation/Booking Status, and View Billing Transactions. These functions allow users to conveniently manage their reservations and monitor their transactions within the system. The diagram emphasizes a streamlined booking experience that reduces manual processing and improves accessibility for customers.

For the administrators and staff, the system provides management and monitoring capabilities necessary for daily operations. Admins can manage reservations, oversee membership status, handle walk-in and advance bookings, manage user accounts, and monitor billing and sales reports. Through the View Sales Report**s** feature, administrators can review summarized transaction records to support operational monitoring and decision-making.

Overall, the Use-Case Diagram demonstrates how the proposed system supports efficient reservation management, organized user handling, accurate billing, and sales monitoring. It also highlights the role-based structure of the system, ensuring that each actor is provided with appropriate access and functions to maintain security, efficiency, and effective study hub management.

3.12 Entity Relationship Diagram (ERD)

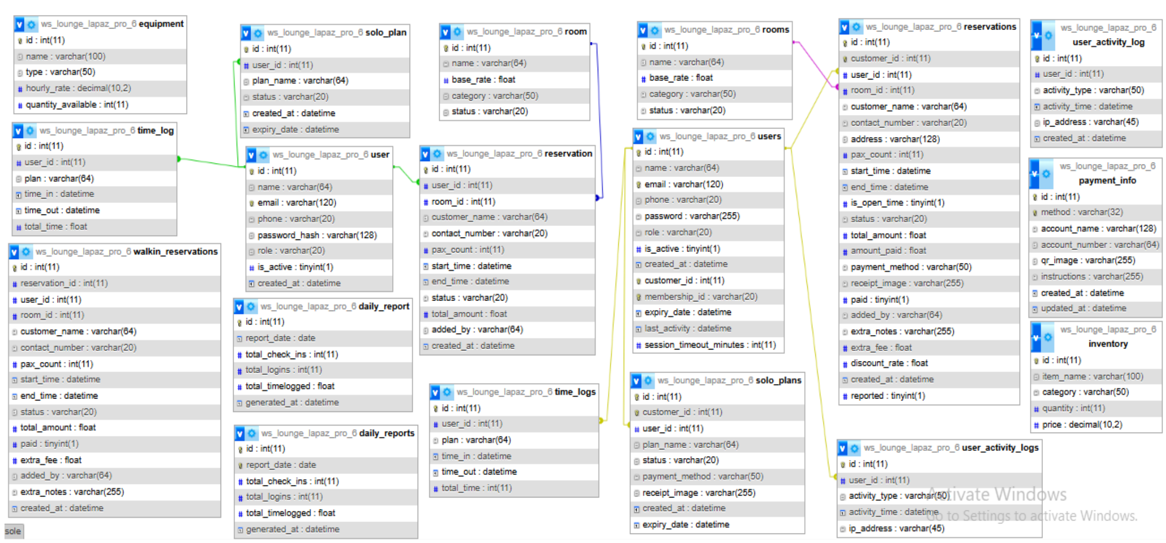


Figure 7. Entity Relationship Diagram of Integrated Study Hub Management System

Significance of Entity Relationship Diagram

The Entity-Relationship Diagram (ERD) serves as the foundational architectural blueprint for the data management layer of the Integrated Study Hub Management System. By providing a visual and structural representation of the relational schema, the ERD allows the researchers to conceptualize how complex business operations—such as multi-tier room reservations, inventory distribution, dynamic time-logging, and digital payments—are translated into organized data entities (Pressman & Maxim, 2020).

The primary significance of this diagram lies in its capacity to enforce structural data integrity and prevent relational anomalies. Through the precise application of cardinal and key constraints, the diagram maps out the direct dependencies between core user activities and security logs, ensuring that no orphan records occur during high-volume transactions. As a Database Design Map, each entity defined in the ERD directly correlates to physical database tables where attributes, primary keys, and foreign keys are explicitly declared to maintain high data consistency (Silberschatz et al., 2019). Furthermore, the ERD acts as a critical validation tool among the proponents and stakeholders, ensuring that the programmatic flow of the application aligns perfectly with the actual operational constraints and administrative workflows of the WS Students & Professionals Lounge.

3.13 Database Design

The database of the Integrated Study Hub Management System is engineered to securely store, organize, and retrieve critical operational metrics, financial transactions, and user histories. The system utilizes a relational database model implemented via MySQL, where data tables are heavily interconnected through Primary Keys (PK) and Foreign Keys (FK) to secure data synchronization (Silberschatz et al., 2019).

The following are the finalized main tables, structural schema, and operational roles derived directly from the system's physical database design:

1. users (ws_lounge_lapaz_pro_6_users)

- **Operational Role:** The core entity that holds demographic, access level, and active status records for all individuals interacting with the hub, including Administrators, Front-desk Staff, and General Customers.
- **Attributes and Schema Specifications:**

id: int(11) **(PK)** — Unique auto-incremented identifier for each user.

name: varchar(64) — Full name of the user.

email: varchar(120) — Registered email address for notifications and logins.

phone: varchar(20) — Active contact number of the user.

password: varchar(255) — Secure, hashed authentication password.

role: varchar(20) — System privilege classification (e.g., Admin, Staff, Customer).

is_active: tinyint(1) — Binary flag for account status validation.

created_at: datetime — Timestamp indicating account registration date.

customer_id: int(11) — External identifier for localized indexing.

membership_id: varchar(20) — Tracks active subscription numbers.

expiry_date: datetime — Deadline for account or membership validity.

last_activity: datetime — Real-time stamp for monitoring active online sessions.

session_timeout_minutes: int(11) — Security configuration variable for session lifespans.

2. reservations (ws_lounge_lapaz_pro_6_reservations)

- **Operational Role:** Captures the comprehensive life cycle of a workspace booking, managing scheduling details, spatial allocation, and advanced billing computations
- **Attributes and Schema Specifications:**

id: int(11) **(PK)** — Unique identifier for each booking instance.

customer_id: int(11) — Primary customer linking constraint.

user_id: int(11) **(FK)** — Relational link to the user table.

room_id: int(11) **(FK)** — Relational link to the specific room booked.

customer_name: varchar(64) — Explicit name label for rapid display queries.

contact_number: varchar(20) — Temporary or reference phone detail.

address: varchar(128) — Physical address background of the client.

pax_count: int(11) — Total headcount utilizing the reserved space.

start_time: datetime — Slotted timestamp for the commencement of the booking.

end_time: datetime — Slotted timestamp for the expiration of the booking.

is_open_time: tinyint(1) — Variable determining flexible or fixed-hour extensions.

status: varchar(20) — Operational state of the booking (e.g., Pending, Confirmed, Cancelled).

total_amount: float — Computed initial charge for space usage.

amount_paid: float — Actual cash/digital value received from the client.

payment_method: varchar(50) — Transaction mode identifier.

receipt_image: varchar(255) — File path or URL registry for digital payment proof uploads.

paid: tinyint(1) — Boolean flag confirming complete payment settlement.

added_by: varchar(64) — Audit trail tracking who encoded the transaction.

extra_notes: varchar(255) — Free-text field for special customer demands.

extra_fee: float — Overtime charges or auxiliary fee additions.

discount_rate: float — Percentages deducted based on dynamic marketing promos or loyalty tiers.

created_at: datetime — System timestamp for record generation.

reported: tinyint(1) — Admin classification flag indicating audit compilation.

3. rooms (ws_lounge_lapaz_pro_6_rooms)

- **Operational Role:** Acts as the primary inventory registry for physical spaces, including individual study pods, general lounges, and corporate meeting rooms.
- **Attributes and Schema Specifications:**

  id: int(11) **(PK)** — Unique index for each room asset.

  name: varchar(64) — Label designating the space name or room number.

  base_rate: float — Standard hourly pricing blueprint for the room.

  category: varchar(50) — Structural classification (e.g., Private, Shared, Group).

  status: varchar(20) — Live availability status indicator (e.g., Available, Occupied, Maintenance).

4. time_logs (ws_lounge_lapaz_pro_6_time_logs)

- **Operational Role:** Functions as an immutable data ledger that tracks exact physical arrivals and exits of users, serving as the raw data feed for automated billing engines.
- **Attributes and Schema Specifications:**

id: int(11) **(PK)** — Sequential tracking code for every log instance.

user_id: int(11) **(FK)** — Links the timestamp directly to a specific user.

plan: varchar(64) — Tracks what specific pricing archetype is active during the duration.

time_in: datetime — Precise clock-in timestamp.

time_out: datetime — Precise clock-out timestamp.

total_time: int(11) — Net computational minutes or hours used for administrative calculation.

5. solo_plans (ws_lounge_lapaz_pro_6_solo_plans)

- **Operational Role:** Manages individualized, non-group workspace subscription plans, tracking user-specific membership duration and active cycles.
- **Attributes and Schema Specifications:**

id: int(11) **(PK)** — Unique primary index for individual plans.

customer_id: int(11) — System indexing code for identification.

user_id: int(11) **(FK)** — Ties the billing package to an authenticated client account.

plan_name: varchar(64) — Name classification of the subscription package.

status: varchar(20) — Current state of the plan validity (e.g., Active, Expired).

payment_method: varchar(50) — Selected channel for payment collection.

receipt_image: varchar(255) — Digital log file for proof of payment verification.

created_at: datetime — Commencement date of the individual plan subscription.

expiry_date: datetime — Mandatory programmatic cutoff date for the subscription.

6. daily_reports (ws_lounge_lapaz_pro_6_daily_reports)

- **Operational Role:** A data warehouse entity that consolidates transaction logs and user entries into high-level business intelligence snapshots for management review.
- **Attributes and Schema Specifications:**

id: int(11) **(PK)** — Unique report sequence index.

report_date: date — Specific operational calendar date being evaluated.

total_check_ins: int(11) — Aggregated count of successful physical entries for the day.

total_logins: int(11) — Total authentication attempts tracked within the web portal interface.

total_timelogged: float — Cumulative number of workspace hours consumed by clients within the specific date.

generated_at: datetime — Microsecond timestamp indicating exact system report generation time.

7. equipment (ws_lounge_lapaz_pro_6_equipment)

- **Operational Role:** Monitors auxiliary tangible assets that can be rented or utilized within the lounge, such as projectors, extension wires, and adapters.
- **Attributes and Schema Specifications:**

id: int(11) **(PK)** — Unique structural ID for every equipment entry.

name: varchar(100) — Label text for the specific machine or item.

type: varchar(50) — Equipment classification tag (e.g., Electronics, Peripheral, Furniture).

hourly_rate: decimal(10,2) — Fixed surcharge cost added when the specific asset is rented out.

quantity_available: int(11) — Operational inventory stock monitoring variable.

8. user_activity_logs (ws_lounge_lapaz_pro_6_user_activity_logs)

- **Operational Role:** Serves as a vital security audit trail that archives user activities to detect system exploitation and ensure internal operational transparency.
- **Attributes and Schema Specifications:**

id: int(11) **(PK)** — Automated security record ID.

user_id: int(11) **(FK)** — Connects the malicious or standard system action to an explicit account.

activity_type: varchar(50) — System action definition tag (e.g., Login Attempt, Reservation Creation, Report Generation).

activity_time: datetime — Definitive date and hour log of the action.

ip_address: varchar(45) — Captures the unique network point location of the device used for the operation.

9. payment_info (ws_lounge_lapaz_pro_6_payment_info)

- **Operational Role:** Manages digital remittance architectures and payment routing channels for customer awareness during checkout phases.
- **Attributes and Schema Specifications:**

id: int(11) **(PK)** — Unique payment channel identifier.

method: varchar(32) — Financial gateway type (e.g., GCash, PayMaya, Bank Transfer).

account_name: varchar(128) — Business register holder name for receipt verification.

account_number: varchar(64) — Explicit account numeric string where cash flows are targeted.

qr_image: varchar(255) — String directory for the dynamic graphic display of scanned payment assets.

instructions: varchar(255) — Explicit step-by-step text guiding the user through the transaction process.

created_at: datetime — Blueprint formulation timestamp.

updated_at: datetime — Track log for adjustments in administrative banking structures.

10. inventory (ws_lounge_lapaz_pro_6_inventory)

- **Operational Role:** Keeps track of consumable internal resources or items sold within the hub (e.g., snacks, beverages, printing paper).
- **Attributes and Schema Specifications:**

id: int(11) **(PK)** — Unique primary catalog index.

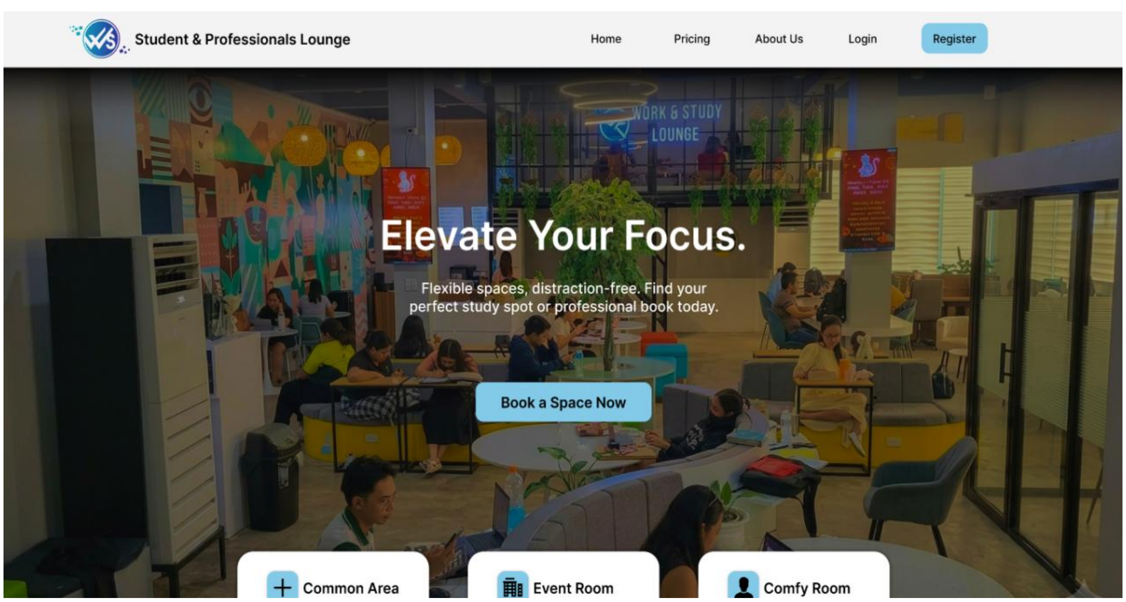
item_name: varchar(100) — Description or name of the physical stock.

category: varchar(50) — Classification category (e.g., Food, Beverage, Office Supplies).

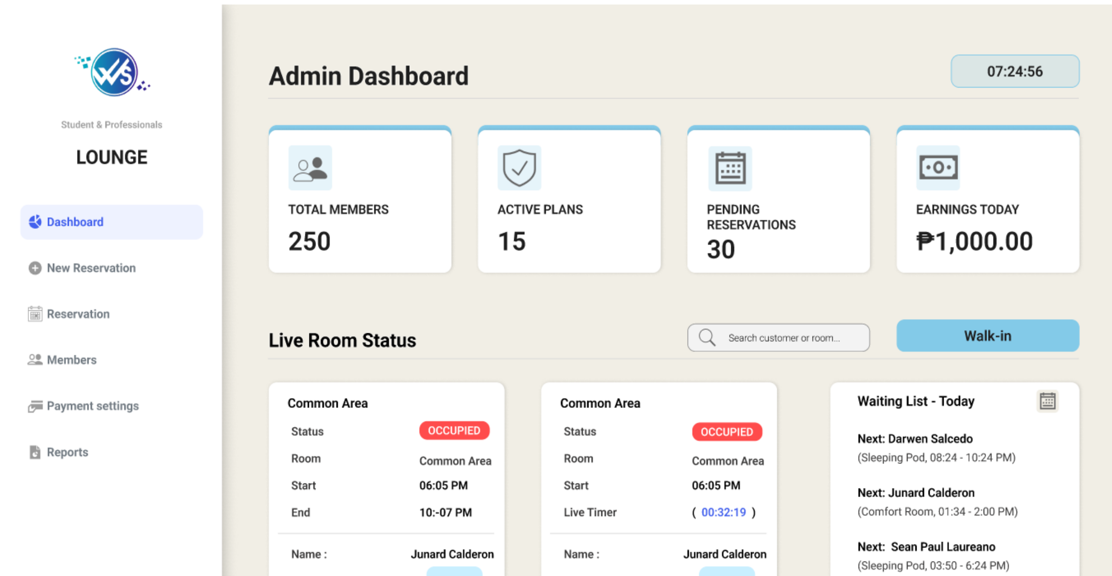
quantity: int(11) — Numerical data tracking remaining units on hand.

price: decimal(10,2) — Operational retail unit value for point-of-sale calculations.

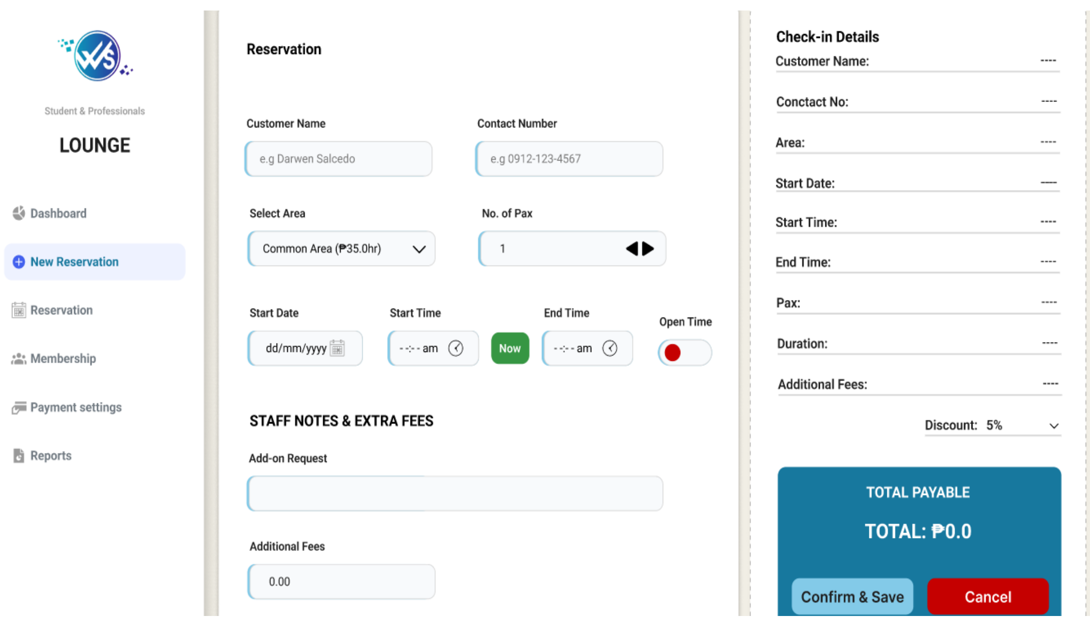
3.14 User Interface Design



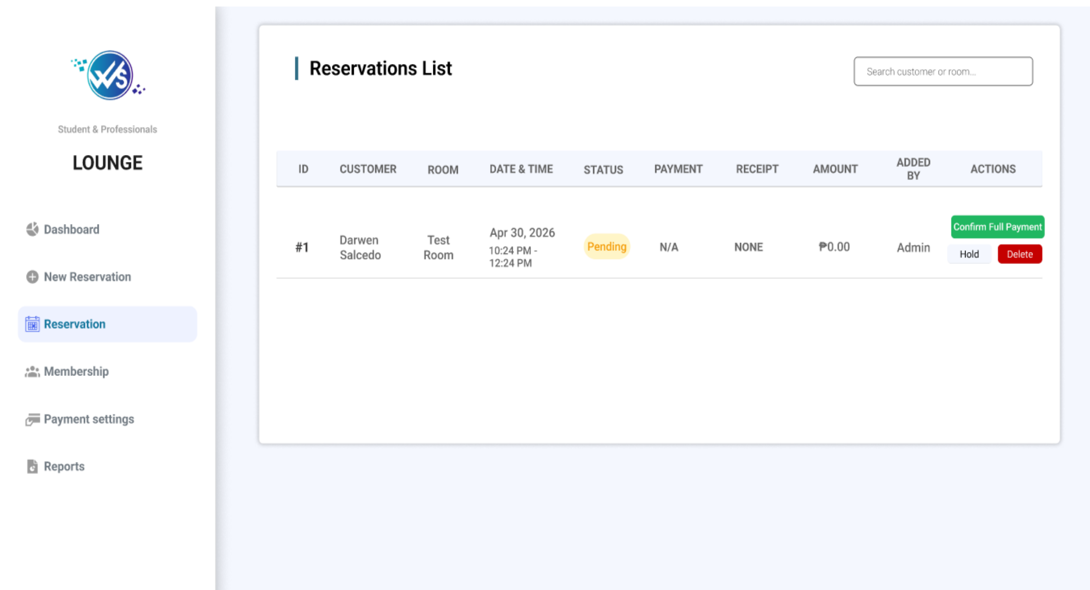
Landing Page



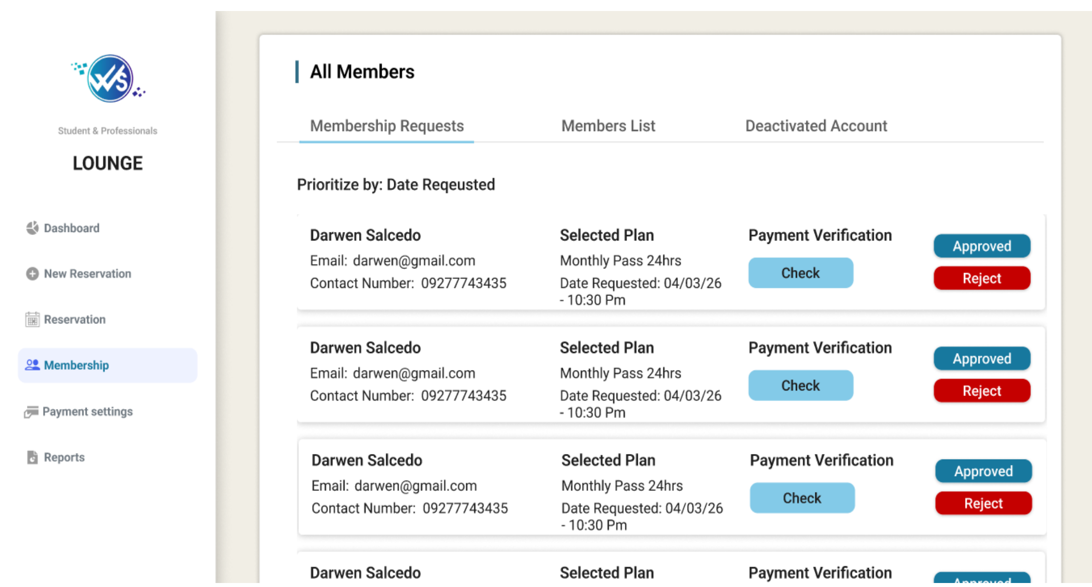
Admin Dashboard



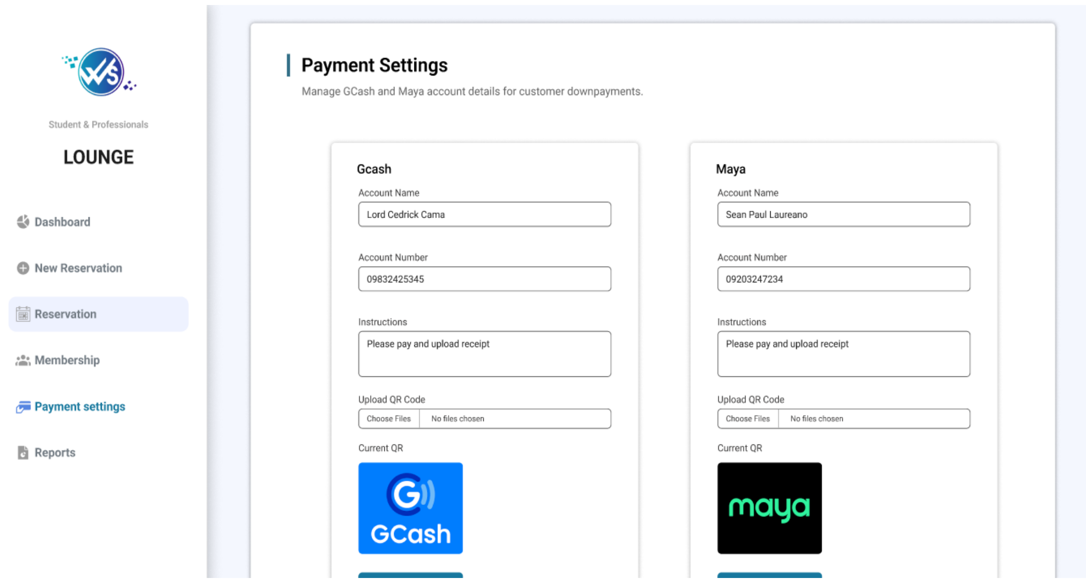
Reservation Bookings



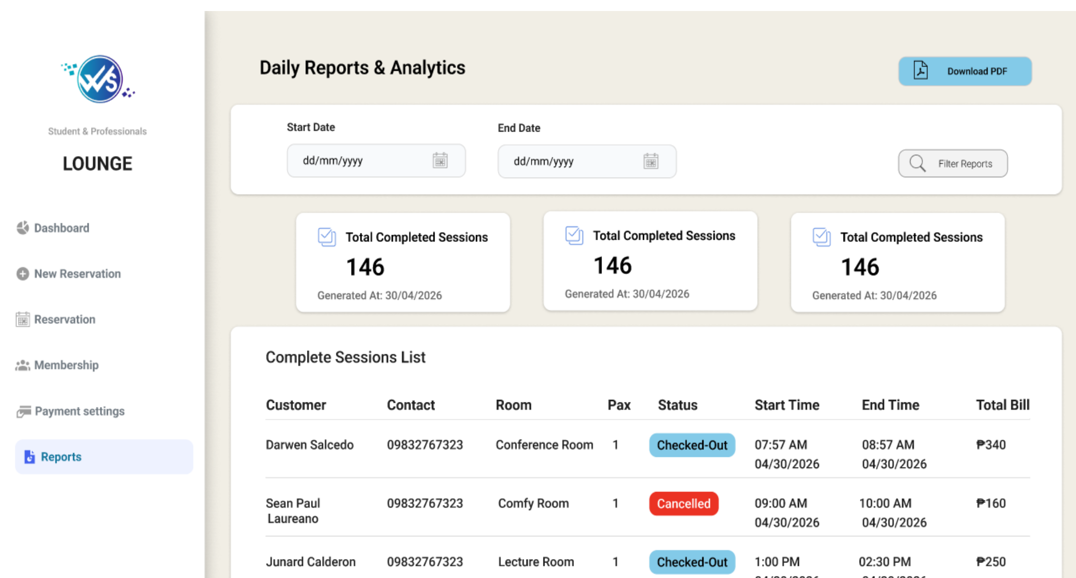
Reservations List



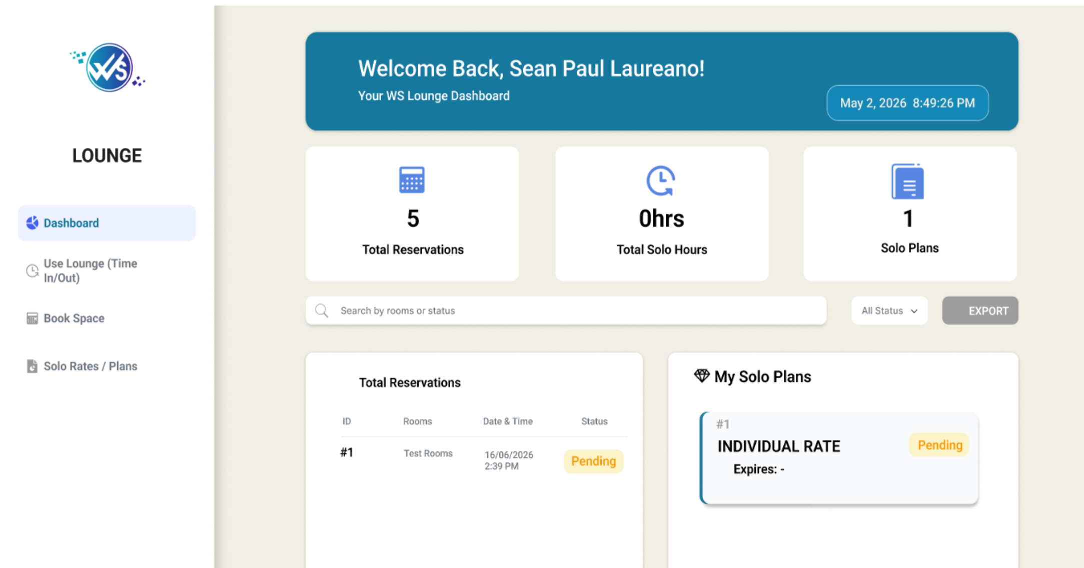
Membership Page



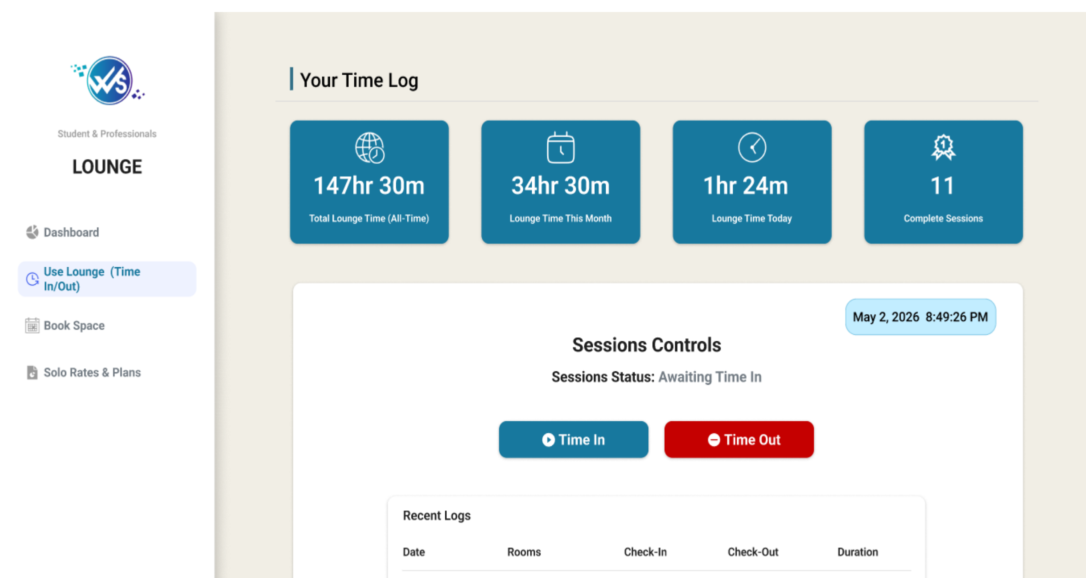
Billing Page



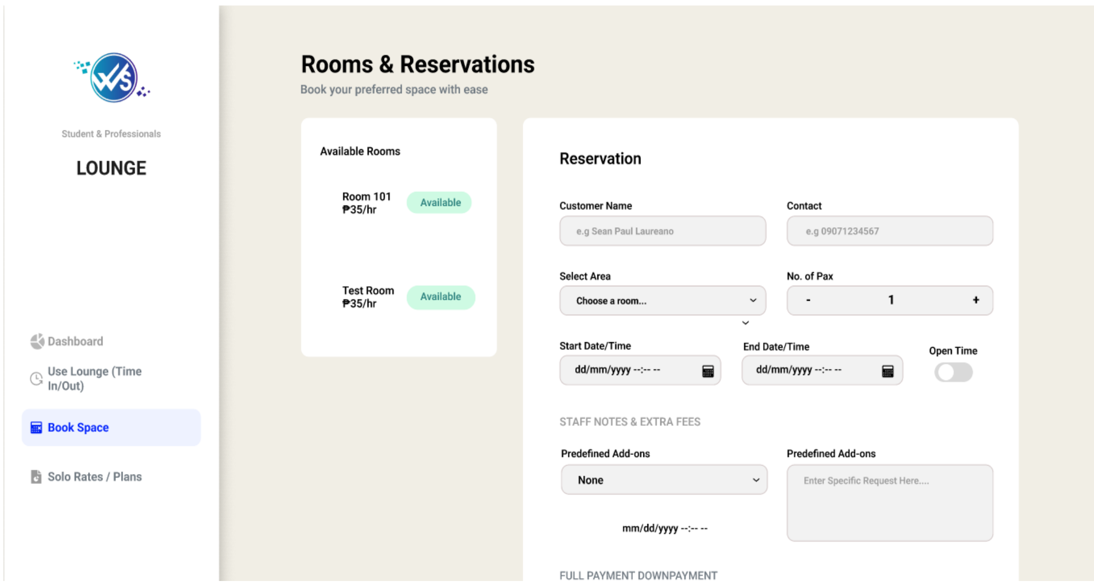
Report History



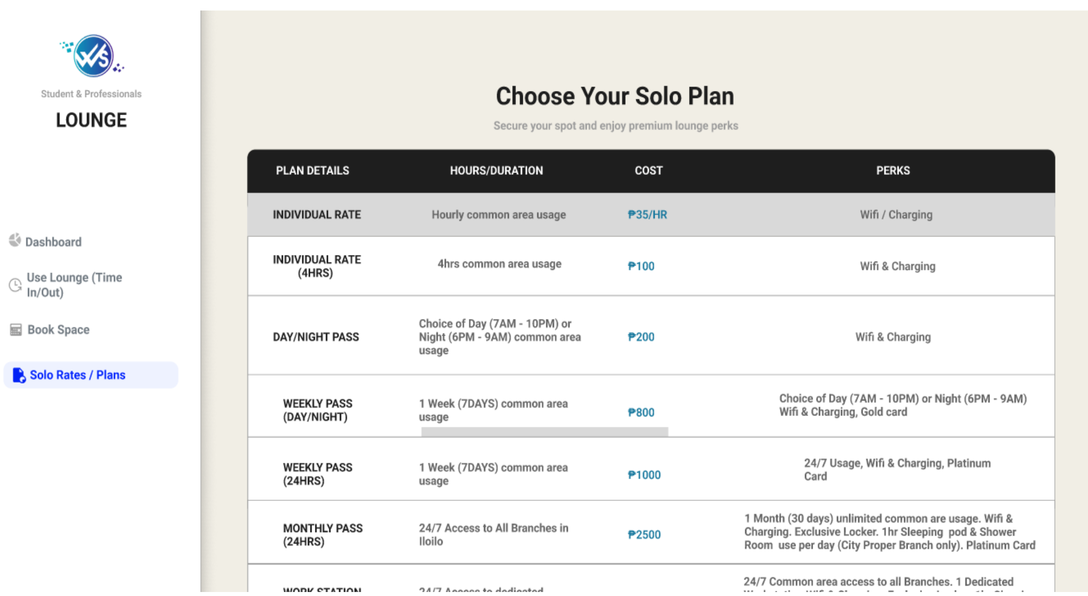
User Dashboard



Time Logs



Reservation Page



Membership Plans

The Integrated Study Hub Management System is composed of several interconnected modules designed to streamline the operations of the WS Students & Professionals Lounge. These components work in harmony to ensure data integrity, security, and a seamless user experience. The system includes the following interfaces:

- **Landing Page:** The digital gateway to the Integrated Study Hub Management System, designed to guide users toward their specific needs—whether they are customers looking for a space or staff members managing operations. It prioritizes clarity and immediate access to core functions.

- **Admin Dashboard:** The central command center for the Integrated Study Hub Management System, providing the branch general manager or owner with total visibility over the lounge's operations. It shifts from the transitional focus of the user to a high-level oversight of financial and operational health.

- **Reservation Bookings:** The system's primary scheduler, designed to manage the life cycle of a study session from the initial request to the physical check-in. It ensures that the WS Students & Professionals Lounge operates at peak efficiency by organizing space allocation and preventing scheduling conflicts.

- **Reservations List:** The core operational tool used by both Staff and Admin to monitor, verify, and manage all scheduled bookings within the hub. It acts as the "master schedule" that ensures the physical lounge areas are organized and utilized efficiently.

- **Membership Page:** The central hub for users to manage their long-term relationship with the lounge. It transitions the user from a one-time visitor to a recurring member, offering structured access to the lounge's premium study environments and professional services

- **Billing Page:** The financial transaction hub of the system, responsible for processing payments, calculating dynamic rates, and issuing digital invoices. It integrates data from the Time Logs and Reservation Bookings to compute precise costs based on room categories, actual hours spent, and applied membership discounts. By maintaining an accurate, real-time ledger of accounts paid, pending fees, and uploaded proof of digital receipts, this module replaces manual computational errors with transparency, ensuring fiscal accountability for both the management and the clients

- **Report History:** The system's permanent archive, allowing the Admin to look back at past performance to identify long-term trends and seasonal patterns. While Daily Reports focus on the immediate "now," the Report History provides the "big picture" necessary for longitudinal business analysis.

- **User Dashboard:** This serves as the primary interface for Students and Professionals, providing a real-time overview of their interactions with the Integrated Study Hub Management System. It is designed to streamline the user experience by consolidating personal activity, financial data, and resource access into a single view.

- **Time Logs:** The primary mechanism for tracking the physical presence and duration of stay for every student and professional in the lounge. It serves as the bridge between a reservation and the final billing process.

- **Reservation Page:** The interactive interface where students and professionals secure their workspace. It is designed to be intuitive, ensuring that the process of booking a study area is fast, transparent, and error-free.

- **Membership Plans:** Represent a tiered subscription model designed to provide consistent value for frequent visitors of the WS Students & Professionals Lounge. By categorizing users into specific plans, the system can automate pricing and access privileges according to the selected tier.

- **Role-Based Access Control -** The primary security protocol that regulates system access, data modification, and operational privileges across the entire platform. By enforcing strict permission boundaries, the system ensures that sensitive business metrics, financial histories, and administrative settings are restricted solely to authorized management roles, while staff and customers are presented only with the specific interfaces necessary for their respective tasks.

Technology Stack

For the development of Integrated Study Hub Management System of WS Students & Professionals Lounge, the proponents use these different programming languages, hardware, software, frameworks, libraries, databases, and different hardware and software tools. The proponents manage the system's security, user-friendly, simple design, performance, and its effectiveness. Below is a more detailed description of the proponents used in creating and designing their system:

3.15 Programming Languages and Tools Used

Programming Languages	Used For
Python	For the main back end language.
SQL	For database queries and schema definition.
HTML/CSS	For frontend templating and styling.
JavaScript	For client-side functionality and interactivity.

Table 2. Programming Languages of an Integrated Study Hub Management System for WS Students & Professionals Lounge

3.15.1 Back-End Frameworks and Libraries

Back-end Frameworks and Library	Purpose or Used for
Flask	The primary web framework is used for routing and request handling.
Flask-SQLAlchemy	Used as an ORM to manage database models and execute queries easily.
Flask-Login	Manages user sessions and authentication for Admin and Staff access.
Flask-WTF / WTForms	Handles secure form processing and provides CSRF protection.
Bcrypt	Used for secure password hashing to protect user credentials.
Flask-SocketIO	Enables real-time, bidirectional communication for live dashboard updates.
MySQL-connector-python	The official driver used to connect the Flask app to the MySQL database.
Flask-Mail	Used for sending automated email notifications to users or administrators.
ReportLab	A library used for generating PDF files and operational reports.

Stripe	Integrated for secure online payment processing for lounge services.
Alembic	Used for database migrations to track changes in the DB schema.
Python-dotenv	Loads environment variables from a .env file for security and configuration.
Werkzeug	Provides WSGI utility functions for the Flask application stack.
Email-validator	Ensures that user-provided email addresses are in the correct format.
Gunicorn	A production-grade WSGI HTTP server used for deploying the application.
Pytest	The framework used for running automated tests to ensure system stability.
Black	A PEP 8 compliant code format used to maintain clean and consistent code.

Table 3. Back-end Frameworks and Libraries of an Integrated Study Hub Management System for WS Students & Professionals Lounge

3.15.2 Front-End Frameworks and Libraries

Front-end Frameworks and Library	**Purpose or Used for**
Vanilla HTML / CSS	Used for the core structure and custom styling of the user interface.
Vanilla JavaScript	Handles client-side interactivity and application logic.
Font Awesome (via CDN)	Provides the icon library used throughout the system’s UI.
Google Fonts (Inter)	The primary typography used for a clean and professional look.
Chart.js (via CDN)	Used for creating visual charts and data visualizations for reports.
Socket.IO (Client)	Handles real-time notifications and live updates on the dashboard.
Icons8	Source of external decorative icons for the system’s statistics and menus.
Moqups	Used for creating wire frames and initial UI designs of the system.
Canva	Used for designing graphic assets, presentations, and branding elements.
Favicon	Used to provide a small, iconic image for the browser tab of the system.

Table 4. Front-End Frameworks and Libraries of an Integrated Study Hub Management System for WS Students & Professionals Lounge

3.15.3 Software Tools

- **Visual Studio Code** – The primary source code editor used for writing and managing the system’s code.
- **GitHub** – For version control and collaborative management of the project's source code.
- **Figma / Canva / Moqups** – Used for creating diagrams, flowcharts, and the final presentation.
- **XAMPP / MySQL Workbench** – For managing the local database and testing queries.

3.15.4 Software Specifications

- **Database:** MySQL 8.0 (or MariaDB)
- **Python Version:** 3.10
- **Web Framework:** Flask 2.x

3.15.5 Hardware Specifications

- **Processor:** Intel(R) Celeron(R) CPU 3867U @ 1.80GHz (1.80 GHz)
- **RAM:** 8.00 GB
- **Storage:** 119 GB

3.15.6 Minimum Hardware Requirements for User (Client)

- **Device:** Any smartphone, tablet, laptop, or PC with an active internet connection.
- **Browser:** Google Chrome, Mozilla Firefox, or Microsoft Edge.

3.15.7 Software Requirements

- **Operating System:** Windows 11 Pro, 64-bit os
- **Python:** 3.8+ version.
- **Database Drivers:** MySQL Connector for Python.

3.16 Survey Questions for System Assessment

This survey aims to assess the user requirements and expectations for the "Integrated Study Hub Management System." The data gathered will help the proponents address the general operational challenges within the current manual setup of WS Students & Professionals Lounge. Your honest feedback is essential in developing a more streamlined and effective management solution. All responses will be handled with strict confidentiality.

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

Demographic Information

1. What is your role in WS Students & Professionals Lounge?

- ☐ Manager / Owner
- ☐ Staff / Employee
- ☐ Regular Customer (Student/Professional)
- ☐ Occasional Visitor

If other (Please Specify): ____________

2. How long have you been using study hubs or similar lounge facilities?

- ☐ Less than 1 year
- ☐ 1-2 years
- ☐ 3-4 years
- ☐ 5-6 years
- ☐ More than 6 years

3. What type of user are you?

- ☐ Student
- ☐ Professional / Freelancer
- ☐ Occasional Visitor

4. What method do you currently use for lounge reservations?

☐ Walk-in / Face-to-face
☐ Social Media (Messenger/Instagram)
☐ Phone Call / Text

If other (Please Specify): ____________

5. How satisfied are you with the current reservation/booking method you are using?

☐ Very Satisfied
☐ Satisfied
☐ Neutral
☐ Not Satisfied
☐ Very Not Satisfied

6. What operational challenges do you face with the current manual system? (Select all that apply)

☐ Double Bookings
☐ Inaccurate Time Tracking
☐ Lack of Real-time Availability
☐ Slow Billing Process

If other (Please Specify): ____________

7. How would you rate the overall quality of the current lounge management process?

- ☐ Very High
- ☐ High
- ☐ Average
- ☐ Low
- ☐ Very Low

8. How satisfied are you with the current security and privacy of your records/data?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Not Satisfied
- ☐ Very Not Satisfied

9. How satisfied are you with the speed of confirming your booking?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Not Satisfied
- ☐ Very Not Satisfied

10. How satisfied are you with the design or user interface (UI) of the current manual setup?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Not Satisfied
- ☐ Very Not Satisfied

11. How would you rate the support and service provided for your lounge concerns?

- ☐ Very Good
- ☐ Good
- ☐ Neutral
- ☐ Poor
- ☐ Very Poor

12. How frequently do you encounter issues like records being lost or incorrect time logs?

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Frequent
- ☐ Always

13. How many times do you visit or book the lounge per month?

- ☐ Less than 5
- ☐ 5-10
- ☐ 11-20
- ☐ 21-30
- ☐ More than 30

14. What is your average duration of stay in the lounge?

- ☐ Less than 2 hours
- ☐ 2-5 hours
- ☐ 6-8 hours
- ☐ Whole day
- ☐ More than a day (Membership)

15. How satisfied are you with the current notification system for your bookings?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Not Satisfied
- ☐ Very Not Satisfied

16. What type of services do you frequently avail in the lounge?

- ☐ Private Study Pods
- ☐ Group Meeting Rooms
- ☐ Common Area
- ☐ Food & Beverage Services

If other (Please Specify): ________________

17. What security features do you prefer for your system account? (Select all that apply)

- ☐ Strong Password Policy
- ☐ Role-Based Access
- ☐ Login Notifications
- ☐ Secure Transaction History

18. What type of transaction method have you been using?

- ☐ Face-to-face (Manual)
- ☐ Online Booking
- ☐ Phone-based Reservation
- ☐ Walk-in

19. What type of information do you usually share with the lounge?

- ☐ Contact Details
- ☐ Identification (Student/Professional ID)
- ☐ Payment Information
- ☐ Schedule/Preferences

20. Who do you usually interact with for your lounge needs?

- ☐ Owner
- ☐ Lounge Manager
- ☐ Front Desk Staff
- ☐ Security Personnel

3.17 Sampling Methods for Respondents

To ensure a balanced and comprehensive evaluation of the system, the researchers will implement a stratified sampling technique. This approach guarantees that the diverse user base of WS Students & Professionals Lounge—from the daily customers to the administrative personnel—is accurately represented in the data collection process. The following steps outline the execution of this strategy:

1. **Define Strata:** Participants will be divided into specific subgroups based on their interaction with the lounge. These groups consist of Students, Professionals/Freelancers, Visitors, Front line Staff, and Business Management. Segmenting the respondents this way allows the proponents to analyze distinct user requirements and operational pain points from various viewpoints.

2. **Sampling Within Strata:** A random selection process will be applied within each group to maintain objectivity. For the guest categories (Students and Professionals), the proponents will invite respondents during peak and off-peak hours to gather a wide range of feedback. For the organizational side, the study will involve the actual employees and managers currently handling the lounge's daily transactions.

3. **Target Sample Size:** The proponents intend to secure a minimum of 30 to 50 completed surveys. This sample size is sufficient to identify recurring operational themes and provide a reliable basis for the system's design and feature validation.

4. **Survey Distribution:** The assessment will be conducted through a hybrid approach. An online version of the survey will be deployed via Google Forms, accessible through social media and email. Simultaneously, the researchers will facilitate on-site data gathering using printed questionnaires and tablets for walk-in clients. To ensure maximum participation, the digital survey is optimized for all major platforms, including Android, iOS, and Windows, allowing for seamless access regardless of the respondent's device.

3.18 Data Analysis Techniques

To determine the efficiency and user-responsiveness of the Integrated Study Hub Management System, the researchers will apply a dual approach of quantitative and qualitative data analysis. This framework allows for the conversion of raw information—collected through structured surveys, stakeholder interviews, and technical system logs—into comprehensive findings that evaluate the software's performance and operational impact.

The study will utilize Descriptive Statistics to provide a clear summary of the data, such as identifying the proportion of clients who are satisfied with the existing manual transactions versus those who encounter operational bottlenecks. This method helps the proponents quantify user experiences, such as the frequency of booking errors or the number of hours typically spent by users in the hub.

Furthermore, Cross-tabulation will be used to analyze the correlation between different user demographics, such as comparing the specific requirements and satisfaction ratings of Students against those of Working Professionals. To establish a standard for user feedback, the Mean, Median, and Mode will be calculated, providing an average score for the system's usability and functionality. To enhance the clarity of the report, the proponents will present these findings through visual tools like Bar Graphs and Pie Charts, enabling a straightforward comparison of trends between the lounge's previous manual operations and the newly implemented digital solution.

3.19 Methods for Processing Raw Data

To uphold the accuracy and reliability of the research findings, a systematic protocol for handling raw data will be strictly followed before any formal analysis occurs. Once information is gathered through surveys, client interviews, and operational observations, it will undergo a multi-stage processing phase to ensure data integrity and readiness for interpretation.

1. **Data Scrubbing and Auditing:** The initial stage involves a thorough review of the collected survey responses to identify any gaps, such as incomplete forms or duplicate entries. Any submissions that contain significant errors or inconsistencies will be isolated or excluded to maintain the quality of the data set.

2. **Systematic Coding:** After scrubbing, a coding system will be implemented for both quantitative ratings and open-ended feedback. This involves assigning specific numerical or thematic labels to responses, allowing for a more structured and streamlined classification during the analysis phase.

3. **Data Tabulation and Management:** The researchers will then organize the coded data by migrating it into centralized spreadsheets. This structured format ensures that the information is easily accessible for statistical computation and cross-referencing.

4. **Verification and Validation:** A validation check will be performed to confirm that the digitized data accurately reflects the original input from the participants. This step is crucial to prevent any clerical errors during the manual encoding process.

5. **Final Synthesis and Preparation:** In the final stage, the data will be summarized and formatted into tables or charts. This prepares the findings for a comprehensive evaluation against the system's development goals and user requirements.

3.20 Frameworks for Evaluating System Effectiveness

To ensure that the Integrated Study Hub Management System aligns with the technical requirements and user expectations of the WS Students & Professionals Lounge, the proponents will adopt established evaluation frameworks. These models provide a structured criteria for assessing the platform's performance, security, and overall user acceptance.

The study will utilize the ISO/IEC 25010 Software Quality Model as the primary standard for technical evaluation. This framework allows the researchers to measure the system's quality through key characteristics such as Functional Suitability, Performance Efficiency, Compatibility, Usability, Security, Maintainability, and Portability. By applying this model, the proponents can verify if the system effectively executes its core tasks, operates reliably under varying user loads, and remains compatible with the various hardware and software environments used in the hub.

Furthermore, the Technology Acceptance Model (TAM) will be implemented to gauge the adoption readiness of both the staff and the customers. Through the analysis of user feedback, the proponents will evaluate the Perceived Usefulness (PU) and Perceived

Ease of Use (PEOU) of the system. This assessment will determine if the participants believe the system simplifies the lounge’s transaction flow and if the interface is intuitive enough for daily operation.

The proponents are committed to maintaining transparency and high ethical standards throughout the research and development life cycle. To ensure the protection of all participants, the following protocols will be observed:

- **Voluntary Participation:** The researchers will strictly avoid any form of coercion or exploitation. Participants will be fully informed of the study's nature and can choose to withdraw at any time.

- **Academic and Professional Approval:** Ethical clearance will be sought from the academic review board and the lounge management before the commencement of data gathering to ensure compliance with institutional standards.

- **Confidentiality and Respect:** All contributions from the respondents will be handled with the utmost respect. Personal data will be anonymize, and the findings will be reported with full integrity, ensuring no manipulation or misrepresentation of results.

- **Accountability in Development:** The final system is designed to promote a secure and accountable environment, ensuring that the lounge's operational data is handled responsibly and that the software fosters trust among its users.

3.21 Ethical Considerations

The proponents are committed to upholding the highest standards of research ethics throughout the development and implementation of the Integrated Study Hub Management System. The following ethical principles will serve as the foundation of the study:

- **Commitment to Data Privacy**
  - Implementing secure authentication measures to ensure that sensitive lounge records and personal information are only accessible to authorized personnel.
  - Safeguarding user information against unauthorized access, potential data breaches, and accidental leaks.

- **System Integrity and Security**
  - Applying robust data protection protocols to maintain the confidentiality and accuracy of all financial and operational records.
  - Conducting regular system evaluations to identify and address vulnerabilities, ensuring a safe digital environment for the lounge's transactions.

- **Informed Consent and Transparency**
  -

- Providing clear information regarding the system's functions and data usage policies before users engage with the platform.
- Ensuring that respondents and system users are fully aware of how their activity logs and personal data will be managed and stored.

- **Prevention of System Misuse**
  - Integrating accountability features, such as transaction logs and role-based permissions, to discourage the fraudulent use of the system.
  - Ensuring that the management tools are used strictly for professional and operational purposes within the lounge.

- **Inclusive and Universal Access**
  - Designing an intuitive user interface that is accessible to all students and professionals, regardless of their level of technical expertise.
  - Ensuring the system remains fair and unbiased in its automated pricing and membership features.

- **Compliance with Regulatory Standards**
  - Strict adherence to the Data Privacy Act of 2012 (Republic Act No. 10173) of the Philippines to ensure legal compliance in handling sensitive information.
  - Maintaining ethical standards in data management and administrative reporting.

- **Professional and Ethical Responsibility**

- Demonstrating a commitment to objective and responsible decision-making at every stage of the software development life cycle.
- Prioritizing the welfare of the lounge’s stakeholders through transparent operations and reliable system performance.